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Techniques for cross-carrier scheduling from a secondary cell to a primary cell

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Abstract

Methods, systems, and devices for wireless communications are described. A user equipment (UE) may be configured to monitor search spaces on a primary cell and an secondary cell in a carrier aggregation configuration. The UE may monitor the search spaces for control information according to a shared search space monitoring configuration. Based on the shared search space monitoring configuration, the UE may identify that the secondary cell is a scheduling cell and the primary cell is a scheduled cell (e.g., cross-carrier scheduling). The UE may monitor control channel candidates of search spaces on the primary cell and on the secondary cell for downlink control information messages scheduling data transmissions with the primary cell. In some examples, the cross-carrier scheduled communications may be configured according to a scheduling configuration, which may include one or more scheduling constraints.

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Background/Summary

CROSS-REFERENCE
(1) The present Application is a 371 national stage filing of International PCT Application No. PCT/CN2021/073960 by TAKEDA et al. entitled “TECHNIQUES FOR CROSS-CARRIER SCHEDULING FROM A SECONDARY CELL TO A PRIMARY CELL,” filed Jan. 27, 2021; and claims priority to International Patent Application No. PCT/CN2020/074063 by TAKEDA et al. entitled “TECHNIQUES FOR CROSS-CARRIER SCHEDULING FROM A SECONDARY CELL TO A PRIMARY CELL,” Jan. 29, 2020, each of which is assigned to the assignee hereof, and each of which is expressly incorporated by reference in its entirety herein.
FIELD OF TECHNOLOGY
(2) The following relates generally to wireless communications and more specifically to techniques for cross-carrier scheduling from a secondary cell to a primary cell.
BACKGROUND
(3) Wireless communications systems are widely deployed to provide various types of communication content such as voice, video, packet data, messaging, broadcast, and so on. These systems may be capable of supporting communication with multiple users by sharing the available system resources (e.g., time, frequency, and power). Examples of such multiple-access systems include fourth generation (4G) systems such as Long Term Evolution (LTE) systems, LTE-Advanced (LTE-A) systems, or LTE-A Pro systems, and fifth generation (5G) systems which may be referred to as New Radio (NR) systems. These systems may employ technologies such as code division multiple access (CDMA), time division multiple access (TDMA), frequency division multiple access (FDMA), orthogonal frequency division multiple access (OFDMA), or discrete Fourier transform spread orthogonal frequency division multiplexing (DFT-S-OFDM). A wireless multiple-access communications system may include one or more base stations or one or more network access nodes, each simultaneously supporting communication for multiple communication devices, which may be otherwise known as user equipment (UE).
(4) In some examples, a UE may communicate with multiple serving cells of a wireless communications system in a carrier aggregation (CA) configuration. In some examples, a serving cell may transmit a cross-carrier scheduling indication that the UE is to communicate with a different serving cell via a data transmission. For example, the UE may be scheduled to receive a downlink transmission, or transmit an uplink transmission.
SUMMARY
(5) The described techniques relate to improved methods, systems, devices, and apparatuses that support techniques for cross-carrier scheduling from a secondary cell to a primary cell. Generally, the described techniques provide for enabling a user equipment (UE) to monitor search spaces on a

primary cell and a secondary cell in a carrier aggregation configuration. The UE may monitor search spaces of the primary cell and the secondary cell for control information according to a shared search space monitoring configuration. Based on the shared search space monitoring configuration, the UE may identify that the secondary cell is a scheduling cell and the primary cell is a scheduled cell (e.g., cross-carrier scheduling). The UE may monitor control channel candidates of search spaces on the primary cell and on the secondary cell for downlink control information messages scheduling data transmissions with the primary cell.
(6) In some examples, the cross-carrier scheduled communications may be configured according to a scheduling configuration, which may include one or more scheduling constraints. The scheduling constraints may apply to the downlink control information messages, the scheduled data transmissions, feedback messages corresponding to the data transmissions, or any combination thereof. The scheduling constraints may enable the UE to improve communications efficiency and resource utilization, without significantly increasing communications complexity for the UE.
(7) A method of wireless communications at a UE is described. The method may include identifying that the UE is in communication with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell, monitoring a common search space of the primary cell and a UE-specific search space of the secondary cell for control information in accordance with a shared search space monitoring configuration, receiving one or more downlink control information messages as a result of monitoring the UE-specific search space of the secondary cell where the one or more downlink control information messages include a cross-carrier scheduling indication that a data transmission is scheduled with the primary cell, and communicating with the primary cell via the data transmission in accordance with at least one of the one or more downlink control information messages.
(8) An apparatus for wireless communications at a UE is described. The apparatus may include a processor, memory coupled with the processor, and instructions stored in the memory. The instructions may be executable by the processor to cause the apparatus to identify that the UE is in communication with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell, monitor a common search space of the primary cell and a UE-specific search space of the secondary cell for control information in accordance with a shared search space monitoring configuration, receive one or more downlink control information messages as a result of monitoring the UE-specific search space of the secondary cell, where the one or more downlink control information messages include a cross-carrier scheduling indication that a data transmission is scheduled with the primary cell, and communicate with the primary cell via the data transmission in accordance with at least one of the one or more downlink control information messages.
(9) Another apparatus for wireless communications at a UE is described. The apparatus may include means for identifying that the UE is in communication with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell, monitoring a common search space of the primary cell and a UE-specific search space of the secondary cell for control information in accordance with a shared search space monitoring configuration, receiving one or more downlink control information messages as a result of monitoring the UE-specific search space of the secondary cell, where the one or more downlink control information messages include a cross-carrier scheduling indication that a data transmission is scheduled with the primary cell, and communicating with the primary cell via the data transmission in accordance with at least one of the one or more downlink control information messages.
(10) A non-transitory computer-readable medium storing code for wireless communications at a UE is described. The code may include instructions executable by a processor to identify that the UE is in communication with a primary cell and a secondary cell in a carrier aggregation configuration,

where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell, monitor a common search space of the primary cell and a UE-specific search space of the secondary cell for control information in accordance with a shared search space monitoring configuration, receive one or more downlink control information messages as a result of monitoring the UE-specific search space of the secondary cell, where the one or more downlink control information messages include a cross-carrier scheduling indication that a data transmission is scheduled with the primary cell, and communicate with the primary cell via the data transmission in accordance with at least one of the one or more downlink control information messages.

(11) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for identifying a respective carrier indicator field in each of the one or more downlink control information messages, where the cross-carrier scheduling indication may be a value included in one of the respective carrier indicator fields.

(12) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining a limit corresponding to the shared search space monitoring configuration, where the limit may be associated with a maximum number of blind decodes to be attempted by the UE, a maximum number of non-overlapping control channel elements, or both, across control channel candidates of the common search space of the primary cell and of the UE-specific search space of the secondary cell.

(13) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving an indication of the limit from the secondary cell, where determining the limit may be based on receiving the indication.

(14) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving an indication of the shared search space monitoring configuration consistent with the limit, where the monitoring may be further in accordance with the limit.

(15) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving an indication of the shared search space monitoring configuration, where the shared search space monitoring configuration includes a number of blind decodes that exceeds the maximum number of blind decodes associated with the limit, a number of non-overlapping control channel elements that exceeds the maximum number of non-overlapping control channel elements associated with the limit, or both, identifying a second UE-specific search space of the secondary cell included in the shared search space monitoring configuration, and refraining from monitoring the second UE-specific search space of the secondary cell in accordance with the limit.

(16) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving a second downlink control information message as a result of the monitoring, where the second downlink control information message includes a second scheduling indication that a second data transmission may be scheduled with the secondary cell, and communicating with the secondary cell via the second data transmission in accordance with the second downlink control information message.

(17) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving a third downlink control information message as a result of the monitoring, where the third downlink control information message includes a third scheduling indication that a third data transmission may be scheduled with the primary cell, and communicating with the primary cell via the third data transmission in accordance with the third downlink control information message.

(18) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the shared search space monitoring configuration includes control channel candidates associated with the one or more downlink control information messages that include the cross-carrier scheduling indication, and the control channel candidates associated with the one or more downlink control information messages may be not included in UE-specific search spaces not associated with the cross-carrier scheduling indication.
(19) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the shared search space monitoring configuration includes control channel candidates associated with the one or more downlink control information messages that include the cross-carrier scheduling indication, and one or more of the control channel candidates associated with the one or more downlink control information messages may be included in a second UE-specific search space not associated with the cross-carrier scheduling indication.
(20) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for reporting an indication of a capability of the UE to support shared search space control channel candidates in UE-specific search spaces not associated with the cross-carrier scheduling indication, where the shared search space monitoring configuration may be based on the capability.
(21) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, monitoring the common search space of the primary cell for the control information may include operations, features, means, or instructions for monitoring a type0 common search space of the primary cell, a type0A common search space of the primary cell, a type1 common search space of the primary cell, a type2 common search space of the primary cell, a type3 common search space of the primary cell, or any combination thereof.
(22) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, communicating with the primary cell via the data transmission may include operations, features, means, or instructions for receiving a downlink data transmission from the primary cell, transmitting an uplink data transmission to the primary cell, or both.
(23) A method of wireless communications at a base station is described. The method may include configuring a UE to communicate with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell, configuring a common search space of the primary cell and a UE-specific search space of the secondary cell for transmitting control information in accordance with a shared search space monitoring configuration, transmitting one or more downlink control information messages to the UE in the UE-specific search space of the secondary cell in accordance with the configuring, where the one or more downlink control information messages include a cross-carrier scheduling indication that a data transmission is scheduled with the primary cell, and communicating with the UE via the data transmission of the primary cell in accordance with at least one of the one or more downlink control information messages.
(24) An apparatus for wireless communications at a base station is described. The apparatus may include a processor, memory coupled with the processor, and instructions stored in the memory. The instructions may be executable by the processor to cause the apparatus to configure a UE to communicate with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell, configure a common search space of the primary cell and a UE-specific search space of the secondary cell for transmitting control information in accordance with a shared search space monitoring configuration, transmit one or more downlink control information messages to the UE in the UE-specific search space of the secondary cell in accordance with the configuring, where the one or more downlink control information messages include a cross-carrier scheduling indication that a data transmission is scheduled with the primary cell, and communicate with the UE via the data transmission of the primary cell in accordance with at least one of the one or more downlink

control information messages.
(25) Another apparatus for wireless communications at a base station is described. The apparatus may include means for configuring a UE to communicate with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell, configuring a common search space of the primary cell and a UE-specific search space of the secondary cell for transmitting control information in accordance with a shared search space monitoring configuration, transmitting one or more downlink control information messages to the UE in the UE-specific search space of the secondary cell in accordance with the configuring, where the one or more downlink control information messages include a cross-carrier scheduling indication that a data transmission is scheduled with the primary cell, and communicating with the UE via the data transmission of the primary cell in accordance with at least one of the one or more downlink control information messages.
(26) A non-transitory computer-readable medium storing code for wireless communications at a base station is described. The code may include instructions executable by a processor to configure a UE to communicate with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell, configure a common search space of the primary cell and a UE-specific search space of the secondary cell for transmitting control information in accordance with a shared search space monitoring configuration, transmit one or more downlink control information messages to the UE in the UE-specific search space of the secondary cell in accordance with the configuring, where the one or more downlink control information messages include a cross-carrier scheduling indication that a data transmission is scheduled with the primary cell, and communicate with the UE via the data transmission of the primary cell in accordance with at least one of the one or more downlink control information messages.
(27) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for configuring a respective carrier indicator field in each of the one or more downlink control information messages, where the cross-carrier scheduling indication may be a value included in one of the respective carrier indicator fields.
(28) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining a limit corresponding to the shared search space monitoring configuration, where the limit may be associated with a maximum number of blind decodes to be attempted by the UE, a maximum number of non-overlapping control channel elements, or both, across control channel candidates of the common search space of the primary cell and of the UE-specific search space of the secondary cell.
(29) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for transmitting an indication of the limit to the UE based on determining the limit.
(30) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining the shared search space monitoring configuration based on the limit, where the shared search space monitoring configuration may be consistent with the limit, and transmitting an indication of the shared search space monitoring configuration to the UE.
(31) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining the shared search space monitoring configuration based on the limit, where the shared search space monitoring configuration includes a number of blind decodes that exceeds the maximum number of blind decodes associated with the limit, a number of non-overlapping control channel elements that

exceeds the maximum number of non-overlapping control channel elements associated with the limit, or both, and transmitting an indication of the shared search space monitoring configuration to the UE.
(32) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for transmitting a second downlink control information message, where the second downlink control information message includes a second scheduling indication that a second data transmission may be scheduled with the secondary cell, and communicating with the UE via the second data transmission of the secondary cell in accordance with the second downlink control information message.
(33) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for transmitting a third downlink control information message, where the third downlink control information message includes a third scheduling indication that a third data transmission may be scheduled with the primary cell, and communicating with the UE via the third data transmission of the primary cell in accordance with the third downlink control information message.
(34) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the shared search space monitoring configuration includes control channel candidates associated with the one or more downlink control information messages that include the cross-carrier scheduling indication, and the control channel candidates associated with the one or more downlink control information messages may be not included in UE-specific search spaces not associated with the cross-carrier scheduling indication.
(35) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the shared search space monitoring configuration includes control channel candidates associated with the one or more downlink control information messages that include the cross-carrier scheduling indication, and one or more of the control channel candidates associated with the one or more downlink control information messages may be included in a second UE-specific search space not associated with the cross-carrier scheduling indication.
(36) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving an indication of a capability of the UE to support shared search space control channel candidates in UE-specific search spaces not associated with the cross-carrier scheduling indication, where the shared search space monitoring configuration may be based on receiving the indication.
(37) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the common search space of the primary cell includes a type0 common search space of the primary cell, a type0A common search space of the primary cell, a type1 common search space of the primary cell, a type2 common search space of the primary cell, a type3 common search space of the primary cell, or any combination thereof.
(38) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, communicating with the UE via the data transmission of the primary cell may include operations, features, means, or instructions for transmitting a downlink data transmission to the UE, receiving an uplink data transmission from the UE, or both.
(39) A method of wireless communications at a UE is described. The method may include identifying that the UE is in communication with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell, receiving, from the primary cell, a first downlink control information message that includes a scheduling indication that a first data transmission is scheduled with the primary cell in accordance with a scheduling configuration that satisfies a set of scheduling constraints associated with the primary cell being cross-carrier scheduled by the secondary cell, receiving, from the secondary cell, a second downlink control information message that includes a cross-carrier scheduling indication that a second data transmission is scheduled with

the primary cell in accordance with the scheduling configuration, and communicating with the primary cell via the first data transmission in accordance with the first downlink control information message, and via the second data transmission in accordance with the second downlink control information message.

(40) An apparatus for wireless communications at a UE is described. The apparatus may include a processor, memory coupled with the processor, and instructions stored in the memory. The instructions may be executable by the processor to cause the apparatus to identify that the UE is in communication with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell, receive, from the primary cell, a first downlink control information message that includes a scheduling indication that a first data transmission is scheduled with the primary cell in accordance with a scheduling configuration that satisfies a set of scheduling constraints associated with the primary cell being cross-carrier scheduled by the secondary cell, receive, from the secondary cell, a second downlink control information message that includes a cross-carrier scheduling indication that a second data transmission is scheduled with the primary cell in accordance with the scheduling configuration, and communicate with the primary cell via the first data transmission in accordance with the first downlink control information message, and via the second data transmission in accordance with the second downlink control information message.

(41) Another apparatus for wireless communications at a UE is described. The apparatus may include means for identifying that the UE is in communication with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell, receiving, from the primary cell, a first downlink control information message that includes a scheduling indication that a first data transmission is scheduled with the primary cell in accordance with a scheduling configuration that satisfies a set of scheduling constraints associated with the primary cell being cross-carrier scheduled by the secondary cell, receiving, from the secondary cell, a second downlink control information message that includes a cross-carrier scheduling indication that a second data transmission is scheduled with the primary cell in accordance with the scheduling configuration, and communicating with the primary cell via the first data transmission in accordance with the first downlink control information message, and via the second data transmission in accordance with the second downlink control information message.

(42) A non-transitory computer-readable medium storing code for wireless communications at a UE is described. The code may include instructions executable by a processor to identify that the UE is in communication with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell, receive, from the primary cell, a first downlink control information message that includes a scheduling indication that a first data transmission is scheduled with the primary cell in accordance with a scheduling configuration that satisfies a set of scheduling constraints associated with the primary cell being cross-carrier scheduled by the secondary cell, receive, from the secondary cell, a second downlink control information message that includes a cross-carrier scheduling indication that a second data transmission is scheduled with the primary cell in accordance with the scheduling configuration, and communicate with the primary cell via the first data transmission in accordance with the first downlink control information message, and via the second data transmission in accordance with the second downlink control information message.

(43) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for monitoring a common search space of the primary cell and one or more UE-specific search spaces of the secondary cell for control information in accordance with a shared search space monitoring configuration, where the first downlink control information message and the second downlink control information message may be received as a result of the monitoring, and where the

scheduling configuration may be based on the shared search space monitoring configuration.
(44) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first data transmission may be communicated in a first duration in accordance with the scheduling configuration, and the second data transmission may be communicated in a second duration that does not overlap with the first duration based on one or more scheduling constraints of the set of scheduling constraints.
(45) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first data transmission may be communicated in a first slot in accordance with the scheduling configuration, and the second data transmission may be communicated in a second slot different from the first slot based on one or more scheduling constraints of the set of scheduling constraints.
(46) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first downlink control information message may be received before the second downlink control information message in accordance with the scheduling configuration, and the first data transmission may be communicated before the second data transmission based on one or more scheduling constraints of the set of scheduling constraints.
(47) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first downlink control information message may be received after the second downlink control information message in accordance with the scheduling configuration, and the first data transmission may be communicated after the second data transmission based on one or more scheduling constraints of the set of scheduling constraints.
(48) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first downlink control information message may be received in a first duration in accordance with the scheduling configuration, and the second downlink control information message may be received in a second duration that does not overlap with the first duration based on one or more scheduling constraints of the set of scheduling constraints.
(49) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first downlink control information message may be received in a first monitoring occasion in accordance with the scheduling configuration, and the second downlink control information message may be received in a second monitoring occasion different from the first monitoring occasion based on one or more scheduling constraints of the set of scheduling constraints.
(50) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first downlink control information message may be received in a first slot in accordance with the scheduling configuration, and the second downlink control information message may be received in a second slot different from the first slot based on one or more scheduling constraints of the set of scheduling constraints.
(51) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for transmitting a first feedback message to the primary cell based on communicating with the primary cell via the first data transmission, and transmitting a second feedback message to the primary cell based on communicating with the primary cell via the second data transmission.
(52) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first feedback message may be included in a first uplink transmission in accordance with the scheduling configuration, and the second feedback message may be included in a second uplink transmission different from the first uplink transmission based on one or more scheduling constraints of the set of scheduling constraints.
(53) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first data transmission may be communicated before the second data transmission in accordance with the scheduling configuration, and the first feedback message may

be transmitted before the second feedback message based on one or more scheduling constraints of the set of scheduling constraints.

(54) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first data transmission may be communicated after the second data transmission in accordance with the scheduling configuration, and the first feedback message may be transmitted after the second feedback message based on one or more scheduling constraints of the set of scheduling constraints.

(55) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving, from the primary cell, a third downlink control information message that includes a second scheduling indication that a retransmission of the first data transmission may be scheduled with the primary cell in accordance with the scheduling configuration, and communicating with the primary cell via the retransmission of the first data transmission in accordance with the third downlink control information message.

(56) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving, from the secondary cell, a fourth downlink control information message that includes a second cross-carrier scheduling indication that a retransmission of the second data transmission may be scheduled with the primary cell in accordance with the scheduling configuration, and communicating with the primary cell via the retransmission of the second data transmission in accordance with the fourth downlink control information message.

(57) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first feedback message includes a hybrid automatic repeat request acknowledgment message or a hybrid automatic repeat request negative acknowledgment message, and the second feedback message includes a hybrid automatic repeat request acknowledgment message or a hybrid automatic repeat request negative acknowledgment message.

(58) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, communicating with the primary cell via a data transmission may include operations, features, means, or instructions for receiving a downlink data transmission from the primary cell, transmitting an uplink data transmission to the primary cell, or both.

(59) A method of wireless communications at a base station is described. The method may include configuring a UE to communicate with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell, transmitting a first downlink control information message via the primary cell that includes a scheduling indication that a first data transmission is scheduled with the primary cell in accordance with a scheduling configuration that satisfies a set of scheduling constraints associated with the primary cell being cross-carrier scheduled by the secondary cell, transmitting a second downlink control information message via the secondary cell that includes a cross-carrier scheduling indication that a second data transmission is scheduled with the primary cell in accordance with the scheduling configuration, and communicating with the UE via the first data transmission of the primary cell in accordance with the first downlink control information message, and via the second data transmission of the primary cell in accordance with the second downlink control information message.

(60) An apparatus for wireless communications at a base station is described. The apparatus may include a processor, memory coupled with the processor, and instructions stored in the memory. The instructions may be executable by the processor to cause the apparatus to configure a UE to communicate with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell, transmit a first downlink control information message via the primary cell that includes a scheduling indication that a first data transmission is scheduled with the primary cell in accordance

with a scheduling configuration that satisfies a set of scheduling constraints associated with the primary cell being cross-carrier scheduled by the secondary cell, transmit a second downlink control information message via the secondary cell that includes a cross-carrier scheduling indication that a second data transmission is scheduled with the primary cell in accordance with the scheduling configuration, and communicate with the UE via the first data transmission of the primary cell in accordance with the first downlink control information message, and via the second data transmission of the primary cell in accordance with the second downlink control information message.
(61) Another apparatus for wireless communications at a base station is described. The apparatus may include means for configuring a UE to communicate with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell, transmitting a first downlink control information message via the primary cell that includes a scheduling indication that a first data transmission is scheduled with the primary cell in accordance with a scheduling configuration that satisfies a set of scheduling constraints associated with the primary cell being cross-carrier scheduled by the secondary cell, transmitting a second downlink control information message via the secondary cell that includes a cross-carrier scheduling indication that a second data transmission is scheduled with the primary cell in accordance with the scheduling configuration, and communicating with the UE via the first data transmission of the primary cell in accordance with the first downlink control information message, and via the second data transmission of the primary cell in accordance with the second downlink control information message.
(62) A non-transitory computer-readable medium storing code for wireless communications at a base station is described. The code may include instructions executable by a processor to configure a UE to communicate with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell, transmit a first downlink control information message via the primary cell that includes a scheduling indication that a first data transmission is scheduled with the primary cell in accordance with a scheduling configuration that satisfies a set of scheduling constraints associated with the primary cell being cross-carrier scheduled by the secondary cell, transmit a second downlink control information message via the secondary cell that includes a cross-carrier scheduling indication that a second data transmission is scheduled with the primary cell in accordance with the scheduling configuration, and communicate with the UE via the first data transmission of the primary cell in accordance with the first downlink control information message, and via the second data transmission of the primary cell in accordance with the second downlink control information message.
(63) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for configuring a common search space of the primary cell and one or more UE-specific search spaces of the secondary cell for transmitting control information in accordance with a shared search space monitoring configuration, where the first downlink control information message and the second downlink control information message may be transmitted based on the determining, and where the scheduling configuration may be based on the shared search space monitoring configuration.
(64) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first data transmission may be communicated in a first duration in accordance with the scheduling configuration, and the second data transmission may be communicated in a second duration that does not overlap with the first duration based on one or more scheduling constraints of the set of scheduling constraints.
(65) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first data transmission may be communicated in a first slot in accordance with the scheduling configuration, and the second data transmission may be communicated in a second

slot different from the first slot based on one or more scheduling constraints of the set of scheduling constraints.

(66) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first downlink control information message may be received before the second downlink control information message in accordance with the scheduling configuration, and the first data transmission may be communicated before the second data transmission based on one or more scheduling constraints of the set of scheduling constraints.

(67) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first downlink control information message may be transmitted after the second downlink control information message in accordance with the scheduling configuration, and the first data transmission may be communicated after the second data transmission based on one or more scheduling constraints of the set of scheduling constraints.

(68) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first downlink control information message may be transmitted in a first duration in accordance with the scheduling configuration, and the second downlink control information message may be received in a second duration that does not overlap with the first duration based on one or more scheduling constraints of the set of scheduling constraints.

(69) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first downlink control information message may be transmitted in a first monitoring occasion in accordance with the scheduling configuration, and the second downlink control information message may be transmitted in a second monitoring occasion different from the first monitoring occasion based on one or more scheduling constraints of the set of scheduling constraints.

(70) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first downlink control information message may be transmitted in a first slot in accordance with the scheduling configuration, and the second downlink control information message may be transmitted in a second slot different from the first slot based on one or more scheduling constraints of the set of scheduling constraints.

(71) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving a first feedback message from the UE based on communicating with the UE via the first data transmission of the primary cell, and receiving a second feedback message from the UE based on communicating with the UE via the first data transmission of the primary cell.

(72) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first feedback message may be included in a first uplink transmission in accordance with the scheduling configuration, and the second feedback message may be included in a second uplink transmission different from the first uplink transmission based on one or more scheduling constraints of the set of scheduling constraints.

(73) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first data transmission may be communicated before the second data transmission in accordance with the scheduling configuration, and the first feedback message may be received before the second feedback message based on one or more scheduling constraints of the set of scheduling constraints.

(74) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first data transmission may be communicated after the second data transmission in accordance with the scheduling configuration, and the first feedback message may be received after the second feedback message based on one or more scheduling constraints of the set of scheduling constraints.

(75) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for transmitting a

third downlink control information message via the primary cell that includes a second scheduling indication that a retransmission of the first data transmission may be scheduled with the primary cell in accordance with the scheduling configuration, and communicating with the UE via the retransmission of the first data transmission of the primary cell in accordance with the third downlink control information message.
(76) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for transmitting a fourth downlink control information message via the secondary cell that includes a second cross-carrier scheduling indication that a retransmission of the second data transmission may be scheduled with the primary cell in accordance with the scheduling configuration, and communicating with the UE via the retransmission of the second data transmission of the primary cell in accordance with the fourth downlink control information message.
(77) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the first feedback message includes a hybrid automatic repeat request acknowledgment message or a hybrid automatic repeat request negative acknowledgment message, and the second feedback message includes a hybrid automatic repeat request acknowledgment message or a hybrid automatic repeat request negative acknowledgment message.
(78) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, communicating with the UE via a data transmission of the primary cell may include operations, features, means, or instructions for transmitting a downlink data transmission to the UE, receiving an uplink data transmission from the UE, or both.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIGS. **1** and **2** illustrate examples of a wireless communications system that supports techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure.
(2) FIG. **3** illustrates an example of a transmission scheme that supports techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure.
(3) FIGS. **4**A through **4**C illustrate examples of timing diagrams that support techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure.
(4) FIGS. **5**A and **5**B illustrate examples of timing diagrams that support techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure.
(5) FIGS. **6**A and **6**B illustrate examples of timing diagrams that support techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure.
(6) FIGS. **7** and **8** illustrate examples of process flows that support techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure.
(7) FIGS. **9** and **10** show block diagrams of devices that support techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure.
(8) FIG. **11** shows a block diagram of a communications manager that supports techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure.

(9) FIG. **12** shows a diagram of a system including a device that supports techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure.
(10) FIGS. **13** and **14** show block diagrams of devices that support techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure.
(11) FIG. **15** shows a block diagram of a communications manager that supports techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure.
(12) FIG. **16** shows a diagram of a system including a device that supports techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure.
(13) FIGS. **17** through **20** show flowcharts illustrating methods that support techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

(14) A user equipment (UE) in a wireless communications system may communicate with multiple serving cells (e.g., associated with one or more base stations) in a carrier aggregation (CA) configuration to increase available bandwidth and data rates for the UE. Each serving cell may communicate with the UE on a respective component carrier (CC) in a respective frequency band. Each serving cell may be associated with a radio access technology (RAT) (e.g., a fourth generation (4G) system such as a Long Term Evolution (LTE) system, a fifth generation (5G) system which may be referred to as a New Radio (NR) system, etc.). For example, a UE may communicate with two serving cells in a dual connectivity scheme, where each serving cell may belong to a different RAT or a different cell group within a RAT.
(15) Each serving cell may belong to one or more cell groups, such as a master cell group (MCG), a secondary cell group (SCG), etc. A UE may use radio resources associated with a master node (MN) for communicating with serving cells in an MCG. Similarly, a UE may use radio resources associated with a secondary node (SN) for communicating with serving cells in an SCG. An MCG may include a primary cell (PCell) and one or more secondary cells (SCells). Similarly, an SCG may include a primary secondary cell (PSCell) and one or more SCells. As used herein, a PCell may refer to a PCell of an MCG or a PSCell of an SCG. A UE may establish a connection with the PCell of an MCG (or with the PSCell of an SCG), and then establish connections with SCells to increase radio resources for communicating with a cell group.
(16) In some examples, a UE may receive control information from one cell scheduling a UE to communicate via a data transmission with another cell (i.e., a different cell than the scheduling cell). This scheduling may be referred to as cross-carrier scheduling. For example, a UE may receive a physical downlink control channel (PDCCH) transmission from an SCell scheduling the UE to communicate a data transmission (e.g., a physical downlink shared channel (PDSCH) transmission, a physical uplink shared channel (PUSCH) transmission, etc.) with a PCell (or a PSCell). The PDCCH transmission may include a downlink control information (DCI) message scheduling the data transmission. In some examples, the DCI message may include a carrier indicator field (CIF) indicating cross-carrier scheduling of the PCell. In some examples, the PDCCH transmission may schedule the UE to communicate a second data transmission with the SCell (i.e., the scheduling cell) in addition to scheduling the data transmission with the PCell.
(17) Cells may transmit control information in a control region such as a control resource set (CORESET) of a system bandwidth (e.g., to limit the bandwidth over which a UE monitors for control information). For example, a CORESET may include one or more search spaces (e.g., a common search space (CSS) and/or a UE-specific search space (USS)) for the transmission of common and UE-specific control information, respectively. A UE may monitor decoding

candidates (e.g., PDCCH candidates) of a CSS for control information intended for multiple UEs as well as decoding candidates of a USS for control information designated for the UE.

(18) In some examples, a UE may be configured to monitor PDCCH candidates of search spaces on an SCell (e.g., a scheduling cell) for cross-carrier scheduling indications associated with transmissions on a PCell (e.g., a scheduled cell). In some examples, the UE may fail to monitor PDCCH candidates of search spaces on the PCell because the UE may expect to receive control information from the SCell. As a result, the UE may miss control information associated with operations configured by the PCell, such as connectivity or mobility functionalities (e.g., fallback operations, PDCCH ordering configurations, beam failure recovery operations, semi-persistent scheduling or configured grant scheduling activation or deactivation, etc.). Missing control information on the PCell may accordingly result in reduced efficiency or reduced reliability in wireless communications at the UE.

(19) According to the techniques described herein, a UE may be configured to monitor search spaces on a PCell and an SCell for control information according to a shared search space monitoring configuration. The UE may monitor PDCCH candidates of one or more CSSs on the PCell and of one or more USSs on the SCell for DCI messages scheduling data transmissions with the PCell. In some examples, the SCell may configure which USSs the UE is to monitor based on a value (e.g., 0 or 1) in a CIF of a DCI message. In some examples, the UE may be configured to monitor a limited quantity of non-overlapping control channel elements (CCEs) of the search spaces to reduce monitoring complexity for the UE. Additionally or alternatively, the UE may be limited to attempting a configured quantity of blind decodes of PDCCH candidates. The limit may be on a per-CC basis. In some examples, the shared search space monitoring configuration may be based on the configured limit. For example, the UE may report a monitoring capability to the cells, and the SCell may update the shared search space monitoring configuration based on the reported monitoring capability. The shared search space monitoring configuration may enable the UE to improve efficiency and reliability of communications with the cells without significantly increasing signaling overhead or monitoring complexity.

(20) In some examples, the cross-carrier scheduled communications may be configured according to a scheduling configuration, which may include one or more scheduling constraints. The scheduling constraints may apply to the DCI messages, the scheduled data transmissions, feedback messages corresponding to the data transmissions, or any combination thereof. For example, data transmissions may be constrained such that they are not time-overlapped, are not scheduled in a same slot, or are not in a different order than the scheduling DCI messages. Similarly, the DCI messages may be constrained such that they are not time-overlapped, are not transmitted in a same monitoring occasion, or are not transmitted in a same slot. The feedback messages corresponding to the data transmissions may be constrained such that they are not transmitted in a same uplink transmission (e.g., a physical uplink control channel (PUCCH) transmission, a PUSCH transmission, etc.), are not in a different order than the corresponding data transmissions, or that a retransmission (e.g., based on a negative acknowledgement (NACK) feedback message) is scheduled by a DCI on the same cell as the original transmission. The scheduling constraints may enable the UE to improve communications efficiency and resource utilization, without significantly increasing communications complexity for the UE.

(21) Aspects of the disclosure are initially described in the context of wireless communications systems. An example transmission scheme, example timing diagrams, and example process flows illustrating aspects of the discussed techniques are then described. Aspects of the disclosure are further illustrated by and described with reference to apparatus diagrams, system diagrams, and flowcharts that relate to techniques for cross-carrier scheduling from a secondary cell to a primary cell.

(22) FIG. **1** illustrates an example of a wireless communications system **100** that supports techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with

aspects of the present disclosure. The wireless communications system **100** may include one or more base stations **105**, one or more UEs **115**, and a core network **130**. In some examples, the wireless communications system **100** may be a Long Term Evolution (LTE) network, an LTE-Advanced (LTE-A) network, an LTE-A Pro network, or a New Radio (NR) network. In some examples, the wireless communications system **100** may support enhanced broadband communications, ultra-reliable (e.g., mission critical) communications, low latency communications, communications with low-cost and low-complexity devices, or any combination thereof.

(23) The base stations **105** may be dispersed throughout a geographic area to form the wireless communications system **100** and may be devices in different forms or having different capabilities. The base stations **105** and the UEs **115** may wirelessly communicate via one or more communication links **125**. Each base station **105** may provide a coverage area **110** over which the UEs **115** and the base station **105** may establish one or more communication links **125**. The coverage area **110** may be an example of a geographic area over which a base station **105** and a UE may support the communication of signals according to one or more radio access technologies.

(24) The UEs **115** may be dispersed throughout a coverage area **110** of the wireless communications system **100**, and each UE may be stationary, or mobile, or both at different times. The UEs **115** may be devices in different forms or having different capabilities. Some example UEs **115** are illustrated in FIG. **1**. The UEs **115** described herein may be able to communicate with various types of devices, such as other UEs **115**, the base stations **105**, or network equipment (e.g., core network nodes, relay devices, integrated access and backhaul (IAB) nodes, or other network equipment), as shown in FIG. **1**.

(25) The base stations **105** may communicate with the core network **130**, or with one another, or both. For example, the base stations **105** may interface with the core network **130** through one or more backhaul links **120** (e.g., via an S1, N2, N3, or other interface). The base stations **105** may communicate with one another over the backhaul links **120** (e.g., via an X2, Xn, or other interface) either directly (e.g., directly between base stations **105**), or indirectly (e.g., via core network **130**), or both. In some examples, the backhaul links **120** may be or include one or more wireless links.

(26) One or more of the base stations **105** described herein may include or may be referred to by a person having ordinary skill in the art as a base transceiver station, a radio base station, an access point, a radio transceiver, a NodeB, an eNodeB (eNB), a next-generation NodeB or a giga-NodeB (either of which may be referred to as a gNB), a Home NodeB, a Home eNodeB, or other suitable terminology.

(27) A UE may include or may be referred to as a mobile device, a wireless device, a remote device, a handheld device, or a subscriber device, or some other suitable terminology, where the “device” may also be referred to as a unit, a station, a terminal, or a client, among other examples. A UE may also include or may be referred to as a personal electronic device such as a cellular phone, a personal digital assistant (PDA), a tablet computer, a laptop computer, or a personal computer. In some examples, a UE may include or be referred to as a wireless local loop (WLL) station, an Internet of Things (IoT) device, an Internet of Everything (IoE) device, or a machine type communications (MTC) device, among other examples, which may be implemented in various objects such as appliances, or vehicles, meters, among other examples.

(28) The UEs **115** described herein may be able to communicate with various types of devices, such as other UEs **115** that may sometimes act as relays as well as the base stations **105** and the network equipment including macro eNBs or gNBs, small cell eNBs or gNBs, or relay base stations, among other examples, as shown in FIG. **1**.

(29) The UEs **115** and the base stations **105** may wirelessly communicate with one another via one or more communication links **125** over one or more carriers. The term “carrier” may refer to a set of radio frequency spectrum resources having a defined physical layer structure for supporting the communication links **125**. For example, a carrier used for a communication link **125** may include a

portion of a radio frequency spectrum band (e.g., a bandwidth part (BWP)) that is operated according to one or more physical layer channels for a given radio access technology (e.g., LTE, LTE-A, LTE-A Pro, NR). Each physical layer channel may carry acquisition signaling (e.g., synchronization signals, system information), control signaling that coordinates operation for the carrier, user data, or other signaling. The wireless communications system **100** may support communication with a UE using carrier aggregation or multi-carrier operation. A UE may be configured with multiple downlink component carriers and one or more uplink component carriers according to a carrier aggregation configuration. Carrier aggregation may be used with both frequency division duplexing (FDD) and time division duplexing (TDD) component carriers.

(30) In some examples (e.g., in a carrier aggregation configuration), a carrier may also have acquisition signaling or control signaling that coordinates operations for other carriers. A carrier may be associated with a frequency channel (e.g., an evolved universal mobile telecommunication system terrestrial radio access (E-UTRA) absolute radio frequency channel number (EARFCN)) and may be positioned according to a channel raster for discovery by the UEs **115**. A carrier may be operated in a standalone mode where initial acquisition and connection may be conducted by the UEs **115** via the carrier, or the carrier may be operated in a non-standalone mode where a connection is anchored using a different carrier (e.g., of the same or a different radio access technology).

(31) The communication links **125** shown in the wireless communications system **100** may include uplink transmissions from a UE to a base station **105**, or downlink transmissions from a base station **105** to a UE. Carriers may carry downlink or uplink communications (e.g., in an FDD mode) or may be configured to carry downlink and uplink communications (e.g., in a TDD mode).

(32) A carrier may be associated with a particular bandwidth of the radio frequency spectrum, and in some examples the carrier bandwidth may be referred to as a “system bandwidth” of the carrier or the wireless communications system **100**. For example, the carrier bandwidth may be one of a number of determined bandwidths for carriers of a particular radio access technology (e.g., 1.4, 3, 5, 10, 15, 20, 40, or 80 megahertz (MHz)). Devices of the wireless communications system **100** (e.g., the base stations **105**, the UEs **115**, or both) may have hardware configurations that support communications over a particular carrier bandwidth or may be configurable to support communications over one of a set of carrier bandwidths. In some examples, the wireless communications system **100** may include base stations **105** or UEs **115** that support simultaneous communications via carriers associated with multiple carrier bandwidths. In some examples, each served UE may be configured for operating over portions (e.g., a sub-band, a BWP) or all of a carrier bandwidth.

(33) Signal waveforms transmitted over a carrier may be made up of multiple subcarriers (e.g., using multi-carrier modulation (MCM) techniques such as orthogonal frequency division multiplexing (OFDM) or discrete Fourier transform spread OFDM (DFT-S-OFDM)). In a system employing MCM techniques, a resource element may consist of one symbol period (e.g., a duration of one modulation symbol) and one subcarrier, where the symbol period and subcarrier spacing are inversely related. The number of bits carried by each resource element may depend on the modulation scheme (e.g., the order of the modulation scheme, the coding rate of the modulation scheme, or both). Thus, the more resource elements that a UE receives and the higher the order of the modulation scheme, the higher the data rate may be for the UE. A wireless communications resource may refer to a combination of a radio frequency spectrum resource, a time resource, and a spatial resource (e.g., spatial layers or beams), and the use of multiple spatial layers may further increase the data rate or data integrity for communications with a UE.

(34) One or more numerologies for a carrier may be supported, where a numerology may include a subcarrier spacing (Δf) and a cyclic prefix. A carrier may be divided into one or more BWPs having the same or different numerologies. In some examples, a UE may be configured with multiple BWPs. In some examples, a single BWP for a carrier may be active at a given time and

communications for the UE may be restricted to one or more active BWPs.
(35) The time intervals for the base stations **105** or the UEs **115** may be expressed in multiples of a basic time unit which may, for example, refer to a sampling period of T.sub.s=1/(Δf.sub.max.Math.N.sub.f) seconds, where Δf.sub.max may represent the maximum supported subcarrier spacing, and N.sub.f may represent the maximum supported discrete Fourier transform (DFT) size. Time intervals of a communications resource may be organized according to radio frames each having a specified duration (e.g., 10 milliseconds (ms)). Each radio frame may be identified by a system frame number (SFN) (e.g., ranging from 0 to 1023).
(36) Each frame may include multiple consecutively numbered subframes or slots, and each subframe or slot may have the same duration. In some examples, a frame may be divided (e.g., in the time domain) into subframes, and each subframe may be further divided into a number of slots. Alternatively, each frame may include a variable number of slots, and the number of slots may depend on subcarrier spacing. Each slot may include a number of symbol periods (e.g., depending on the length of the cyclic prefix prepended to each symbol period). In some wireless communications systems **100**, a slot may further be divided into multiple mini-slots containing one or more symbols. Excluding the cyclic prefix, each symbol period may contain one or more (e.g., N.sub.f) sampling periods. The duration of a symbol period may depend on the subcarrier spacing or frequency band of operation.
(37) A subframe, a slot, a mini-slot, or a symbol may be the smallest scheduling unit (e.g., in the time domain) of the wireless communications system **100** and may be referred to as a transmission time interval (TTI). In some examples, the TTI duration (e.g., the number of symbol periods in a TTI) may be variable. Additionally or alternatively, the smallest scheduling unit of the wireless communications system **100** may be dynamically selected (e.g., in bursts of shortened TTIs (sTTIs)).
(38) Physical channels may be multiplexed on a carrier according to various techniques. A physical control channel and a physical data channel may be multiplexed on a downlink carrier, for example, using one or more of time division multiplexing (TDM) techniques, frequency division multiplexing (FDM) techniques, or hybrid TDM-FDM techniques. A control region (e.g., a CORESET) for a physical control channel may be defined by a number of symbol periods and may extend across the system bandwidth or a subset of the system bandwidth of the carrier. One or more control regions (e.g., CORESETs) may be configured for a set of the UEs **115**. For example, one or more of the UEs **115** may monitor or search control regions for control information according to one or more search space sets, and each search space set may include one or multiple control channel candidates in one or more aggregation levels arranged in a cascaded manner. An aggregation level for a control channel candidate may refer to a number of control channel resources (e.g., control channel elements (CCEs)) associated with encoded information for a control information format having a given payload size. Search space sets may include common search space sets configured for sending control information to multiple UEs **115** and UE-specific search space sets for sending control information to a specific UE.
(39) Each base station **105** may provide communication coverage via one or more cells, for example a macro cell, a small cell, a hot spot, or other types of cells, or any combination thereof. The term “cell” may refer to a logical communication entity used for communication with a base station **105** (e.g., over a carrier) and may be associated with an identifier for distinguishing neighboring cells (e.g., a physical cell identifier (PCID), a virtual cell identifier (VCID), or others). In some examples, a cell may also refer to a geographic coverage area **110** or a portion of a geographic coverage area **110** (e.g., a sector) over which the logical communication entity operates. Such cells may range from smaller areas (e.g., a structure, a subset of structure) to larger areas depending on various factors such as the capabilities of the base station **105**. For example, a cell may be or include a building, a subset of a building, or exterior spaces between or overlapping with geographic coverage areas **110**, among other examples.

(40) A macro cell generally covers a relatively large geographic area (e.g., several kilometers in radius) and may allow unrestricted access by the UEs **115** with service subscriptions with the network provider supporting the macro cell. A small cell may be associated with a lower-powered base station **105**, as compared with a macro cell, and a small cell may operate in the same or different (e.g., licensed, unlicensed) frequency bands as macro cells. Small cells may provide unrestricted access to the UEs **115** with service subscriptions with the network provider or may provide restricted access to the UEs **115** having an association with the small cell (e.g., the UEs **115** in a closed subscriber group (CSG), the UEs **115** associated with users in a home or office). A base station **105** may support one or multiple cells and may also support communications over the one or more cells using one or multiple component carriers.
(41) In some examples, a carrier may support multiple cells, and different cells may be configured according to different protocol types (e.g., MTC, narrowband IoT (NB-IoT), enhanced mobile broadband (eMBB)) that may provide access for different types of devices.
(42) In some examples, a base station **105** may be movable and therefore provide communication coverage for a moving geographic coverage area **110**. In some examples, different geographic coverage areas **110** associated with different technologies may overlap, but the different geographic coverage areas **110** may be supported by the same base station **105**. In other examples, the overlapping geographic coverage areas **110** associated with different technologies may be supported by different base stations **105**. The wireless communications system **100** may include, for example, a heterogeneous network in which different types of the base stations **105** provide coverage for various geographic coverage areas **110** using the same or different radio access technologies.
(43) The wireless communications system **100** may be configured to support ultra-reliable communications or low-latency communications, or various combinations thereof. For example, the wireless communications system **100** may be configured to support ultra-reliable low-latency communications (URLLC) or mission critical communications. The UEs **115** may be designed to support ultra-reliable, low-latency, or critical functions (e.g., mission critical functions). Ultra-reliable communications may include private communication or group communication and may be supported by one or more mission critical services such as mission critical push-to-talk (MCPTT), mission critical video (MCVideo), or mission critical data (MCData). Support for mission critical functions may include prioritization of services, and mission critical services may be used for public safety or general commercial applications. The terms ultra-reliable, low-latency, mission critical, and ultra-reliable low-latency may be used interchangeably herein.
(44) In some examples, a UE may also be able to communicate directly with other UEs **115** over a device-to-device (D2D) communication link **135** (e.g., using a peer-to-peer (P2P) or D2D protocol). One or more UEs **115** utilizing D2D communications may be within the geographic coverage area **110** of a base station **105**. Other UEs **115** in such a group may be outside the geographic coverage area **110** of a base station **105** or be otherwise unable to receive transmissions from a base station **105**. In some examples, groups of the UEs **115** communicating via D2D communications may utilize a one-to-many (1:M) system in which each UE transmits to every other UE in the group. In some examples, a base station **105** facilitates the scheduling of resources for D2D communications. In other cases, D2D communications are carried out between the UEs **115** without the involvement of a base station **105**.
(45) The core network **130** may provide user authentication, access authorization, tracking, Internet Protocol (IP) connectivity, and other access, routing, or mobility functions. The core network **130** may be an evolved packet core (EPC) or 5G core (5GC), which may include at least one control plane entity that manages access and mobility (e.g., a mobility management entity (MME), an access and mobility management function (AMF)) and at least one user plane entity that routes packets or interconnects to external networks (e.g., a serving gateway (S-GW), a Packet Data Network (PDN) gateway (P-GW), or a user plane function (UPF)). The control plane entity may

manage non-access stratum (NAS) functions such as mobility, authentication, and bearer management for the UEs **115** served by the base stations **105** associated with the core network **130**. User IP packets may be transferred through the user plane entity, which may provide IP address allocation as well as other functions. The user plane entity may be connected to the network operators IP services **150**. The operators IP services **150** may include access to the Internet, Intranet(s), an IP Multimedia Subsystem (IMS), or a Packet-Switched Streaming Service.

(46) Some of the network devices, such as a base station **105**, may include subcomponents such as an access network entity **140**, which may be an example of an access node controller (ANC). Each access network entity **140** may communicate with the UEs **115** through one or more other access network transmission entities **145**, which may be referred to as radio heads, smart radio heads, or transmission/reception points (TRPs). Each access network transmission entity **145** may include one or more antenna panels. In some configurations, various functions of each access network entity **140** or base station **105** may be distributed across various network devices (e.g., radio heads and ANCs) or consolidated into a single network device (e.g., a base station **105**).

(47) The wireless communications system **100** may operate using one or more frequency bands, typically in the range of 300 megahertz (MHz) to 300 gigahertz (GHz). Generally, the region from 300 MHz to 3 GHz is known as the ultra-high frequency (UHF) region or decimeter band because the wavelengths range from approximately one decimeter to one meter in length. The UHF waves may be blocked or redirected by buildings and environmental features, but the waves may penetrate structures sufficiently for a macro cell to provide service to the UEs **115** located indoors. The transmission of UHF waves may be associated with smaller antennas and shorter ranges (e.g., less than 100 kilometers) compared to transmission using the smaller frequencies and longer waves of the high frequency (HF) or very high frequency (VHF) portion of the spectrum below 300 MHz.

(48) The wireless communications system **100** may utilize both licensed and unlicensed radio frequency spectrum bands. For example, the wireless communications system **100** may employ License Assisted Access (LAA), LTE-Unlicensed (LTE-U) radio access technology, or NR technology in an unlicensed band such as the 5 GHz industrial, scientific, and medical (ISM) band. When operating in unlicensed radio frequency spectrum bands, devices such as the base stations **105** and the UEs **115** may employ carrier sensing for collision detection and avoidance. In some examples, operations in unlicensed bands may be based on a carrier aggregation configuration in conjunction with component carriers operating in a licensed band (e.g., LAA). Operations in unlicensed spectrum may include downlink transmissions, uplink transmissions, P2P transmissions, or D2D transmissions, among other examples.

(49) A base station **105** or a UE may be equipped with multiple antennas, which may be used to employ techniques such as transmit diversity, receive diversity, multiple-input multiple-output (MIMO) communications, or beamforming. The antennas of a base station **105** or a UE may be located within one or more antenna arrays or antenna panels, which may support MIMO operations or transmit or receive beamforming. For example, one or more base station antennas or antenna arrays may be co-located at an antenna assembly, such as an antenna tower. In some examples, antennas or antenna arrays associated with a base station **105** may be located in diverse geographic locations. A base station **105** may have an antenna array with a number of rows and columns of antenna ports that the base station **105** may use to support beamforming of communications with a UE. Likewise, a UE may have one or more antenna arrays that may support various MIMO or beamforming operations. Additionally or alternatively, an antenna panel may support radio frequency beamforming for a signal transmitted via an antenna port.

(50) Beamforming, which may also be referred to as spatial filtering, directional transmission, or directional reception, is a signal processing technique that may be used at a transmitting device or a receiving device (e.g., a base station **105**, a UE) to shape or steer an antenna beam (e.g., a transmit beam, a receive beam) along a spatial path between the transmitting device and the receiving device. Beamforming may be achieved by combining the signals communicated via antenna

elements of an antenna array such that some signals propagating at particular orientations with respect to an antenna array experience constructive interference while others experience destructive interference. The adjustment of signals communicated via the antenna elements may include a transmitting device or a receiving device applying amplitude offsets, phase offsets, or both to signals carried via the antenna elements associated with the device. The adjustments associated with each of the antenna elements may be defined by a beamforming weight set associated with a particular orientation (e.g., with respect to the antenna array of the transmitting device or receiving device, or with respect to some other orientation).

(51) The wireless communications system **100** may be a packet-based network that operates according to a layered protocol stack. In the user plane, communications at the bearer or Packet Data Convergence Protocol (PDCP) layer may be IP-based. A Radio Link Control (RLC) layer may perform packet segmentation and reassembly to communicate over logical channels. A Medium Access Control (MAC) layer may perform priority handling and multiplexing of logical channels into transport channels. The MAC layer may also use error detection techniques, error correction techniques, or both to support retransmissions at the MAC layer to improve link efficiency. In the control plane, the Radio Resource Control (RRC) protocol layer may provide establishment, configuration, and maintenance of an RRC connection between a UE and a base station **105** or a core network **130** supporting radio bearers for user plane data. At the physical layer, transport channels may be mapped to physical channels.

(52) The UEs **115** and the base stations **105** may support retransmissions of data to increase the likelihood that data is received successfully. Hybrid automatic repeat request (HARQ) feedback is one technique for increasing the likelihood that data is received correctly over a communication link **125**. HARQ may include a combination of error detection (e.g., using a cyclic redundancy check (CRC)), forward error correction (FEC), and retransmission (e.g., automatic repeat request (ARQ)). HARQ may improve throughput at the MAC layer in poor radio conditions (e.g., low signal-to-noise conditions). In some examples, a device may support same-slot HARQ feedback, where the device may provide HARQ feedback in a specific slot for data received in a previous symbol in the slot. In other cases, the device may provide HARQ feedback in a subsequent slot, or according to some other time interval.

(53) A UE may be configured to monitor search spaces on a PCell and an SCell (e.g., of one or more base stations **105**) for control information according to a shared search space monitoring configuration. The UE may monitor PDCCH candidates of one or more CSSs on the PCell and of one or more USSs on the SCell for DCI messages scheduling data transmissions with the PCell. In some examples, the SCell may configure which USSs the UE is to monitor based on a value (e.g., 0 or 1) in a CIF of a DCI message. In some examples, the UE may be configured to monitor a limited quantity of non-overlapping CCEs of the search spaces to reduce monitoring complexity for the UE. Additionally or alternatively, the UE may be limited to attempting a configured quantity of blind decodes of PDCCH candidates. The limit may be on a per-CC basis. In some examples, the shared search space monitoring configuration may be based on the configured limit. For example, the UE may report a monitoring capability to the cells, and the SCell may update the shared search space monitoring configuration based on the reported monitoring capability. The shared search space monitoring configuration may enable the UE to improve efficiency and reliability of communications with the cells without significantly increasing signaling overhead or monitoring complexity.

(54) In some examples, the cross-carrier scheduled communications may be configured according to a scheduling configuration, which may include one or more scheduling constraints. The scheduling constraints may apply to the DCI messages, the scheduled data transmissions, feedback messages corresponding to the data transmissions, or any combination thereof. For example, data transmissions may be constrained such that they are not time-overlapped, are not scheduled in a same slot, or are not in a different order than the scheduling DCI messages. Similarly, the DCI

messages may be constrained such that they are not time-overlapped, are not transmitted in a same monitoring occasion, or are not transmitted in a same slot. The feedback messages corresponding to the data transmissions may be constrained such that they are not transmitted in a same uplink transmission (e.g., a PUCCH transmission, a PUSCH transmission, etc.), are not in a different order than the corresponding data transmissions, or that a retransmission (e.g., based on a HARQ NACK feedback message) is scheduled by a DCI on the same cell as the original transmission. The scheduling constraints may enable the UE to improve communications efficiency and resource utilization, without significantly increasing communications complexity for the UE.

(55) FIG. **2** illustrates an example of a wireless communications system **200** that supports techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure. In some examples, the wireless communications system **200** may implement aspects of wireless communication system **100**. For example, the wireless communications system **200** may include a UE **215**, which may be an example of the corresponding device described with reference to FIG. **1**. The wireless communications system **200** may also include a PCell **205** and an SCell **206**, where the PCell **205** and the SCell **206** may be associated with one or more base stations **105** described with reference to FIG. **1**. In some examples, the PCell **205** and the SCell **206** may be associated with a same base station **105**. In other examples, the PCell **205** may be associated with a first base station **105** and the SCell **206** may be associated with a second base station **105**. The PCell **205** and the SCell **206** may belong to an MCG. In some examples, the UE **215** may be in communication with the SCell **206** and a PSCell (not shown) of an SCG. The wireless communications system **200** may include features for improved cross-carrier scheduling in CA configurations, among other benefits.

(56) In the wireless communications system **200**, the PCell **205** may act as a serving cell for a geographic coverage area **210**-*a*, and the SCell **206** may act as a serving cell for a geographic coverage area **210**-*b*. The PCell **205** and the SCell **206** may configure and transmit DCI messages **220** to the UE **215**. The UE **215** may communicate data transmissions **225** (e.g., PDSCH transmissions, PUSCH transmissions, etc.) with the PCell **205** or the SCell **206**. As illustrated in FIG. **2**, the UE **215** may communicate concurrently with the PCell **205** and the SCell **206**, for example in a CA configuration. The UE **215** may monitor PDCCH candidates of one or more CSSs on the PCell **205** and of one or more USSs on the SCell **206** according to a shared search space monitoring configuration. Based on the shared search space monitoring configuration, the UE **215** may identify that the SCell **206** is a scheduling cell and the PCell **205** is a scheduled cell (e.g., cross-carrier scheduling).

(57) Based on monitoring the PDCCH candidates, the UE **215** may receive a DCI message **220**-*b* from the SCell **206** scheduling the UE **215** to communicate a data transmission **225**-*a* with the PCell **205**. In some examples, the DCI message **220**-*b* may include a CIF indicating cross-carrier scheduling of the PCell **205**. In some examples, a DCI message **220** (e.g., a DCI message **220**-*a* from the PCell **205** or a DCI message **220**-*b* from the SCell **206**) may schedule the UE **215** to communicate additional data transmissions **225** (e.g., a data transmission **225**-*a*, a data transmission **225**-*b*, etc.) with the PCell **205** or the SCell **206**. In some examples, the DCI message **220**-*b* may schedule both the data transmission **225**-*a* and the data transmission **225**-*b* (e.g., joint scheduling).

(58) In some examples, the SCell **206** may configure which USSs the UE **215** is to monitor based on a value (e.g., 0 or 1) in a CIF of the DCI message **220**-*b*. The UE **215** may be configured to monitor a limited quantity of non-overlapping CCEs of the search spaces to reduce monitoring complexity for the UE **215**. Additionally or alternatively, the UE **215** may be limited to attempting a configured quantity of blind decodes of PDCCH candidates. The limit may be on a per-CC basis. In some examples, the shared search space monitoring configuration may be based on the configured limit. In some examples, the SCell **206** may transmit to the UE **215** a shared search space monitoring configuration that configures the UE to perform a quantity of blind decodes or

monitor a quantity of CCEs that exceeds the limit. In such examples, the UE **215** may determine to not monitor one or more USSs in the SCell **206**, which may reduce monitoring complexity. In some examples, the UE **215** may report a monitoring capability to the PCell **205** and the SCell **206**, and the SCell **206** may update the shared search space monitoring configuration based on the reported monitoring capability of the UE **215**. The shared search space monitoring configuration may enable the UE **215** to improve efficiency and reliability of communications with the PCell **205** and the SCell **206** without significantly increasing signaling overhead or monitoring complexity.

(59) FIG. **3** illustrates an example of a transmission scheme **300** that supports techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure. In some examples, the transmission scheme **300** may implement aspects of wireless communication systems **100** and **200**. For example, the transmission scheme **300** may be associated with communications between a PCell **305**, an SCell **306**, and a UE, which may be examples of corresponding devices described with reference to FIGS. **1** and **2**. The PCell **305** and the SCell **306** may belong to an MCG. In some examples, the UE may be in communication with the SCell **306** and a PSCell (not shown) of an SCG. The transmission scheme **300** may illustrate features for improved cross-carrier scheduling in CA configurations, among other benefits.

(60) A UE may communicate concurrently with the PCell **305** and the SCell **306**, for example in a CA configuration. The UE may monitor CSS PDCCH candidates **310** of one or more CSSs on the PCell **305** and USS PDCCH candidates **320** of one or more USSs on the SCell **306** according to a shared search space monitoring configuration. Based on the shared search space monitoring configuration, the UE may identify that the SCell **306** is a scheduling cell and the PCell **305** is a scheduled cell (e.g., cross-carrier scheduling).

(61) As illustrated in FIG. **3**, the CSS PDCCH candidates **310** may be included in a group **315**, and the USS PDCCH candidates **320** may be included in groups **325**. The group **315** may include CSS PDCCH candidates **310** of one or more CSSs, and each group **325** may include USS PDCCH candidates **320** of one or more USSs. Although the group **315** illustrated in FIG. **3** includes CSS PDCCH candidates **310**-*a* through **310**-*d*, the group **315** may in some examples include more or less than four CSS PDCCH candidates **310** based on a configuration of the PCell **305**. Similarly, although the group **325**-*a* includes USS PDCCH candidates **320**-*a* through **320**-*d* and the group **325**-*b* includes USS PDCCH candidates **320**-*e* through **320**-*h*, each group **325** may in some examples include more or less than four USS PDCCH candidates **320** based on a configuration of the SCell **306**. The CSSs on the PCell **305** may include type0-CSSs, type0A-CSSs, type1-CSSs, type2-CSSs, or type3-CSSs, or any combination thereof. Each type of CSS may include different types of signaling (e.g., a system information block (SIB), random access channel (RACH) procedure messages, paging messages, etc.) which the UE may monitor for on the PCell **305** to enable connectivity or mobility functionalities.

(62) In some examples, the UE may be configured with a shared search space monitoring configuration. Based on the shared search space monitoring configuration, a data transmission **330**-*a* (e.g., a PDSCH transmission from the PCell **305**, a PUSCH transmission to the PCell **305**, etc.) on the PCell **305** may be scheduled by a DCI message in a CSS PDCCH candidate **310** on the PCell **305**, or by a DCI message in a USS PDCCH candidate **320** on the SCell **306**, or both. The shared search space monitoring configuration may enable transmission diversity, where a network may schedule the data transmission **330**-*a* using the PCell **305** or the SCell **306** based on network conditions. The shared search space monitoring configuration may also increase a quantity of PDCCH candidates (e.g., CSS PDCCH candidates **310** or USS PDCCH candidates **320**) the UE monitors to receive control information, which may lead to an increase in monitoring or decoding complexity for the UE.

(63) The UE may be configured to monitor a limited quantity of non-overlapping CCEs of the search spaces to reduce monitoring complexity for the UE. Additionally or alternatively, the UE may be limited to attempting a configured quantity of blind decodes of PDCCH candidates. The

limit may be on a per-CC basis. In some examples, the shared search space monitoring configuration may be based on the configured limit. In some examples, the SCell **306** may transmit to the UE a shared search space monitoring configuration that configures the UE to perform a quantity of blind decodes or monitor a quantity of CCEs that exceeds the limit. In such examples, the UE may determine to not monitor one or more USSs in the SCell **306**, which may reduce monitoring complexity. In some examples, the UE may report a monitoring capability to the PCell **305** and the SCell **306**, and the SCell **306** may update the shared search space monitoring configuration based on the reported monitoring capability of the UE.

(64) In some examples, the shared search space monitoring configuration may indicate which USSs the UE is to monitor based on a value (e.g., 0 or 1) in a CIF of a DCI message received in a USS PDCCH candidate **320**. Each group **325** may correspond to a value in the CIF. For example, the group **325-***a* may include USS PDCCH candidates **320** of USSs the UE is to monitor when the value in the CIF is 0, and the group **325-***b* may include USS PDCCH candidates **320** of USSs the UE is to monitor when the value in the CIF is 1.

(65) In a first example of the shared search space monitoring configuration, a first value (e.g., 0) in the CIF may indicate that the USS PDCCH candidates **320** corresponding to the first value in the CIF (e.g., the group **325-***a*) are configured to transmit cross-carrier scheduling indications (e.g., DCI messages scheduling the data transmission **330-***a*). A second value (e.g., 1) in the CIF may indicate that the USS PDCCH candidates **320** corresponding to the second value in the CIF (e.g., the group **325-***b*) are not configured to transmit cross-carrier scheduling indications, and may not be used to transmit DCI messages scheduling the data transmission **330-***a*. Accordingly, based on the shared search space monitoring configuration of the first example, the UE may monitor USS PDCCH candidates **320** of the group **325-***a* for DCI messages scheduling the data transmission **330-***a*, but skip monitoring USS PDCCH candidates **320** of the group **325-***b*.

(66) In a second example of the shared search space monitoring configuration, the UE may be configured to monitor the group **325-***a* and the group **325-***b* for DCI messages scheduling the data transmission **330-***a*. That is, based on the shared search space monitoring configuration of the second example, any or all of the USS PDCCH candidates **320-***a* through **320-***h* may include DCI messages scheduling the data transmission **330-***a*, and the UE may monitor all of the USS PDCCH candidates **320** for the DCI messages. In some cases, the number of USS PDCCH candidates **320** may be limited based on the limit determined by the UE, which may be reported in the monitoring capability.

(67) In a third example, a UE may be configured to support the shared search space monitoring configuration of the first example or the second example based on reporting a monitoring capability. For example, the UE may signal a searchSpaceSharingCA-DL parameter and/or a searchSpaceSharingCA-UL parameter in capability information signaling to indicate that the UE supports the shared search space monitoring configuration of the first example. Alternatively, the UE may signal a searchSpaceSharingCA-DL-toPCell parameter and/or a searchSpaceSharingCA-UL-toPCell parameter in capability information signaling to indicate that the UE supports the shared search space monitoring configuration of the second example. Based on the capability information signaling from the UE, the SCell **306** may transmit to the UE an update to the shared search space monitoring configuration.

(68) In some examples, the UE may additionally monitor USS PDCCH candidates **320** of the groups **325** for DCI messages scheduling a data transmission **330-***b*. That is, based on the shared search space monitoring configuration, any or all of the USS PDCCH candidates **320-***a* through **320-***h* may include DCI messages scheduling the data transmission **330-***a*, the data transmission **330-***b*, or both.

(69) The shared search space monitoring configuration may enable the UE to improve efficiency and reliability of communications with the PCell **305** and the SCell **306** without significantly increasing signaling overhead or monitoring complexity.

(70) FIGS. **4**A through **4**C illustrate examples of timing diagrams **400** that support techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure. In some examples, the timing diagrams **400** may implement aspects of wireless communication systems **100** and **200**. For example, the timing diagrams **400** may be associated with communications between PCells **405**, SCells **406**, and UEs, which may be examples of corresponding devices described with reference to FIGS. **1** and **2**. The timing diagrams **400** may illustrate features for improved cross-carrier scheduling in CA configurations, among other benefits.

(71) A UE may communicate concurrently with a PCell **405** and an SCell **406**, for example in a CA configuration. The UE may monitor CSS PDCCH candidates **410** of one or more CSSs on the PCell **405** and USS PDCCH candidates **420** of one or more USSs on the SCell **406** according to a shared search space monitoring configuration. Based on the shared search space monitoring configuration, the UE may identify that the SCell **406** is a scheduling cell and the PCell **405** is a scheduled cell (e.g., cross-carrier scheduling). Based on monitoring the PDCCH candidates, the UE may receive one or more DCI messages scheduling one or more data transmissions **430** on the PCell **405**.

(72) In some examples, the cross-carrier scheduled communications on the PCell **405** may be configured according to a scheduling configuration, which may include one or more scheduling constraints applied to the data transmissions **430**, the DCI messages, etc. The scheduling constraints may enable the UE to improve communications efficiency and resource utilization, without significantly increasing communications complexity for the UE.

(73) FIG. **4**A illustrates an example timing diagram **400-***a*. In the timing diagram **400-***a*, a first DCI message in a CSS PDCCH candidate **410-***a* on a PCell **405-***a* may schedule a data transmission **430-***a* on the PCell **405-***a*, and a second DCI message in a USS PDCCH candidate **420-***a* on an SCell **406-***a* may schedule a data transmission **430-***b* on the PCell **405-***a*. The data transmissions **430-***a* and **430-***b* may partially or completely overlap in time, which may increase communications complexity for the UE. For example, communicating the data transmission **430-***a* may include the UE using a first hardware configuration to transmit a PUSCH transmission to the PCell **405-***a* or receive a PDSCH transmission from the PCell **405-***a*, and communicating the data transmission **430-***b* may include the UE using a second hardware configuration different from the first hardware configuration. If the data transmissions **430-***a* and **430-***b* overlap in time, the UE may consume additional power using the first and second hardware configurations concurrently. In some examples, the UE may be unable to use the first and second hardware configurations concurrently, and the UE may fail to communicate one or both of the data transmissions **430-***a* and **430-***b*, which may reduce communications reliability and efficiency.

(74) In some examples, the scheduling configuration for cross-carrier scheduled communications on the PCell **405-***a* may include a scheduling constraint that data transmissions **430** may not overlap in time, which may reduce communications complexity for the UE. That is, the UE may communicate the data transmission **430-***a* in a first duration according to the scheduling configuration, and the UE may communicate the data transmission **430-***b* in a second duration that does not overlap in time with the first duration based on the scheduling constraint.

(75) FIG. **4**B illustrates an example timing diagram **400-***b*. In the timing diagram **400-***b*, a first DCI message in a CSS PDCCH candidate **410-***b* on a PCell **405-***b* may schedule a data transmission **430-***c* on the PCell **405-***b*, and a second DCI message in a USS PDCCH candidate **420-***b* on an SCell **406-***b* may schedule a data transmission **430-***d* on the PCell **405-***b*. Based on the scheduling, the UE may communicate the data transmissions **430-***c* and **430-***d* in a slot **415**, which may increase communications complexity for the UE. For example, communicating the data transmission **430-***c* may include the UE using a first hardware configuration to transmit a PUSCH transmission to the PCell **405-***b* or receive a PDSCH transmission from the PCell **405-***b*, and communicating the data transmission **430-***d* may include the UE using a second hardware configuration different from the

first hardware configuration. If the data transmissions **430**-*c* and **430**-*d* are scheduled in the same slot **415**, the UE may not have sufficient time to transition between the first and second hardware configurations, and the UE may fail to communicate one or both of the data transmissions **430**-*c* and **430**-*d*, which may reduce communications reliability and efficiency.

(76) In some examples, the scheduling configuration for cross-carrier scheduled communications on the PCell **405**-*b* may include a scheduling constraint that data transmissions **430** may not be scheduled in the same slot **415**, which may reduce communications complexity for the UE. That is, the UE may communicate the data transmission **430**-*c* in a first slot (e.g., the slot **415**) according to the scheduling configuration, and the UE may communicate the data transmission **430**-*d* in a second slot different from the first slot (e.g., a slot before or after the slot **415**) based on the scheduling constraint.

(77) FIG. **4**C illustrates an example timing diagram **400**-*c*. In the timing diagram **400**-*c*, a first DCI message in a CSS PDCCH candidate **410**-*c* on a PCell **405**-*c* may schedule a data transmission **430**-*e* on the PCell **405**-*c*, and a second DCI message in a USS PDCCH candidate **420**-*c* on an SCell **406**-*c* may schedule a data transmission **430**-*f* on the PCell **405**-*c*. The SCell **406**-*c* may transmit the USS PDCCH candidate **420**-*c* at a time **425**-*a*, and the PCell **405**-*c* may transmit the CSS PDCCH candidate **410**-*c* at a time **425**-*b* after the time **425**-*a*. The data transmissions **430**-*e* may be scheduled before the data transmission **430**-*f*. That is, the data transmissions **430**-*e* and **430**-*f* may be scheduled in a different order than the order in which the UE receives the scheduling DCI messages, which may increase communications complexity for the UE. For example, communicating the data transmission **430**-*e* may include the UE using a first hardware configuration to transmit a PUSCH transmission to the PCell **405**-*c* or receive a PDSCH transmission from the PCell **405**-*c*, and communicating the data transmission **430**-*f* may include the UE using a second hardware configuration different from the first hardware configuration.

(78) If the data transmissions **430**-*e* and **430**-*f* are scheduled in a different order than the order in which the UE receives the scheduling DCI messages, the UE may consume additional power by performing unnecessary transitions between hardware configurations. For example, the UE may transition to the second hardware configuration upon receiving the second DCI message at the time **425**-*a*. The UE may then transition to the first hardware configuration upon receiving the first DCI message at the time **425**-*b* to communicate the data transmission **430**-*e*, before transitioning back to the second hardware configuration to communicate the data transmission **430**-*f*. In some examples, the UE may not have sufficient time to make additional transitions between the first and second hardware configurations, and the UE may fail to communicate one or both of the data transmissions **430**-*e* and **430**-*f*, which may reduce communications reliability and efficiency.

(79) In some examples, the scheduling configuration for cross-carrier scheduled communications on the PCell **405**-*c* may include a scheduling constraint that data transmissions **430** may not be scheduled in a different order than the order in which the UE receives the scheduling DCI messages, which may reduce communications complexity for the UE. That is, the UE may receive the first DCI message in the CSS PDCCH candidate **410**-*c* after receiving the second DCI message in the USS PDCCH candidate **420**-*c* according to the scheduling configuration, and the UE may be scheduled to communicate the data transmission **430**-*e* after communicating the data transmission **430**-*f* based on the scheduling constraint. Alternatively, the UE may receive the first DCI message in the CSS PDCCH candidate **410**-*c* before receiving the second DCI message in the USS PDCCH candidate **420**-*c* according to the scheduling configuration, and the UE may be scheduled to communicate the data transmission **430**-*e* before communicating the data transmission **430**-*f* based on the scheduling constraint.

(80) FIGS. **5**A and **5**B illustrate examples of timing diagrams **500** that support techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure. In some examples, the timing diagrams **500** may implement aspects of wireless communication systems **100** and **200**. For example, the timing diagrams **500** may be associated

with communications between PCells **505**, SCells **506**, and UEs, which may be examples of corresponding devices described with reference to FIGS. **1** and **2**. The timing diagrams **500** may illustrate features for improved cross-carrier scheduling in CA configurations, among other benefits.
(81) A UE may communicate concurrently with a PCell **505** and an SCell **506**, for example in a CA configuration. The UE may monitor CSS PDCCH candidates **510** of one or more CSSs on the PCell **505** and USS PDCCH candidates **520** of one or more USSs on the SCell **506** according to a shared search space monitoring configuration. Based on the shared search space monitoring configuration, the UE may identify that the SCell **506** is a scheduling cell and the PCell **505** is a scheduled cell (e.g., cross-carrier scheduling). Based on monitoring the PDCCH candidates, the UE may receive one or more DCI messages scheduling one or more data transmissions **530** on the PCell **505**.
(82) In some examples, the cross-carrier scheduled communications on the PCell **505** may be configured according to a scheduling configuration, which may include one or more scheduling constraints applied to transmission of the DCI messages in the PDCCH candidates, the data transmissions **530**, etc. The scheduling constraints may enable the UE to improve communications efficiency and resource utilization, without significantly increasing communications complexity for the UE.
(83) FIG. **5**A illustrates an example timing diagram **500**-*a*. In the timing diagram **500**-*a*, a first DCI message in a CSS PDCCH candidate **510**-*a* on a PCell **505**-*a* may schedule a data transmission **530**-*a* on the PCell **505**-*a*, and a second DCI message in a USS PDCCH candidate **520**-*a* on an SCell **506**-*a* may schedule a data transmission **530**-*b* on the PCell **505**-*a*. The CSS PDCCH candidate **510**-*a* and the USS PDCCH candidate **520**-*a* may partially or completely overlap in time, or the PDCCH candidates may be in a same monitoring occasion **525**-*a*, which may increase monitoring complexity for the UE. For example, the UE may use a first hardware configuration to monitor the CSS PDCCH candidate **510**-*a* and a second hardware configuration different from the first hardware configuration to monitor the USS PDCCH candidate **520**-*a*. Additionally or alternatively, the UE may be limited in the number of blind decodes the UE may attempt across PDCCH candidates based on a shared search space monitoring configuration as described herein. If the CSS PDCCH candidate **510**-*a* and the USS PDCCH candidate **520**-*a* containing the DCI messages overlap in time, or are in the same monitoring occasion **525**-*a*, the UE may miss one or both of the DCI messages. Based on missing the DCI messages, the UE may fail to communicate one or both of the data transmissions **530**-*a* and **530**-*b*, which may reduce communications reliability and efficiency.
(84) In some examples, the scheduling configuration for cross-carrier scheduled communications on the PCell **505**-*a* may include a scheduling constraint that PDCCH candidates scheduling data transmissions **530** may not overlap in time or be in a same monitoring occasion **525**, which may reduce communications complexity for the UE. That is, the PCell **505**-*a* may transmit the first DCI message in a CSS PDCCH candidate **510**-*a* in a first duration according to the scheduling configuration, and the SCell **506**-*a* may transmit the second DCI message in a USS PDCCH candidate **520**-*a* in a second duration that does not overlap in time with the first duration or is not in the same monitoring occasion **525** as the first duration based on the scheduling constraint.
(85) FIG. **5**B illustrates an example timing diagram **500**-*b*. In the timing diagram **500**-*b*, a first DCI message in a CSS PDCCH candidate **510**-*b* on a PCell **505**-*b* may schedule a data transmission **530**-*c* on the PCell **505**-*b*, and a second DCI message in a USS PDCCH candidate **520**-*b* on an SCell **506**-*b* may schedule a data transmission **530**-*d* on the PCell **505**-*b*. The CSS PDCCH candidate **510**-*b* may be in a monitoring occasion **525**-*b* in a slot **515**, and the USS PDCCH candidate **520**-*b* may be in a monitoring occasion **525**-*c* in the same slot **515**, which may increase monitoring complexity for the UE. For example, the UE may use a first hardware configuration to monitor the CSS PDCCH candidate **510**-*b* and a second hardware configuration different from the

first hardware configuration to monitor the USS PDCCH candidate **520**-*b*. Additionally or alternatively, the UE may be limited in the number of blind decodes the UE may attempt across PDCCH candidates based on a shared search space monitoring configuration as described herein. If the CSS PDCCH candidate **510**-*b* and the USS PDCCH candidate **520**-*b* are in the same slot **515**, the UE may miss one or both of the DCI messages. Based on missing the DCI messages, the UE may fail to communicate one or both of the data transmissions **530**-*c* and **530**-*d*, which may reduce communications reliability and efficiency.

(86) In some examples, the scheduling configuration for cross-carrier scheduled communications on the PCell **505**-*b* may include a scheduling constraint that PDCCH candidates scheduling data transmissions **530** may not be in a same slot **515**, which may reduce communications complexity for the UE. That is, the PCell **505**-*b* may transmit the first DCI message in a CSS PDCCH candidate **510**-*b* in a first slot (e.g., the slot **515**) according to the scheduling configuration, and the SCell **506**-*b* may transmit the second DCI message in a USS PDCCH candidate **520**-*b* in a second slot different from the first slot (e.g., in a slot before or after the slot **515**) based on the scheduling constraint.

(87) FIGS. **6**A and **6**B illustrate examples of timing diagrams **600** that support techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure. In some examples, the timing diagrams **600** may implement aspects of wireless communication systems **100** and **200**. For example, the timing diagrams **600** may be associated with communications between PCells **605**, SCells **606**, and UEs, which may be examples of corresponding devices described with reference to FIGS. **1** and **2**. The timing diagrams **600** may illustrate features for improved cross-carrier scheduling in CA configurations, among other benefits.

(88) A UE may communicate concurrently with a PCell **605** and an SCell **606**, for example in a CA configuration. The UE may monitor CSS PDCCH candidates **610** of one or more CSSs on the PCell **605** and USS PDCCH candidates **620** of one or more USSs on the SCell **606** according to a shared search space monitoring configuration. Based on the shared search space monitoring configuration, the UE may identify that the SCell **606** is a scheduling cell and the PCell **605** is a scheduled cell (e.g., cross-carrier scheduling). Based on monitoring the PDCCH candidates, the UE may receive one or more DCI messages scheduling one or more downlink transmissions **630** (e.g., PDSCH transmissions) from the PCell **605** to the UE. In some examples, the UE may transmit feedback messages (e.g., HARQ-ACK or HARQ-NACK feedback) corresponding to the downlink transmissions **630** in uplink transmissions **635** (e.g., PUSCH transmissions, PDCCH transmissions, etc.).

(89) In some examples, the cross-carrier scheduled communications on the PCell **605** may be configured according to a scheduling configuration, which may include one or more scheduling constraints applied to the downlink transmissions **630**, the feedback messages, etc. The scheduling constraints may enable the UE to improve communications efficiency and resource utilization, without significantly increasing communications complexity for the UE.

(90) FIG. **6**A illustrates an example timing diagram **600**-*a*. In the timing diagram **600**-*a*, a first DCI message in a CSS PDCCH candidate **610**-*a* on a PCell **605**-*a* may schedule a downlink transmission **630**-*a* on the PCell **605**-*a*, and a second DCI message in a USS PDCCH candidate **620**-*a* on an SCell **606**-*a* may schedule a downlink transmission **630**-*b* on the PCell **605**-*a*. The UE may report a first feedback message corresponding to the downlink transmission **630**-*a* and a second feedback message corresponding to the downlink transmission **630**-*b*. The UE may report both the first and second feedback messages in an uplink transmission **635**-*a*, which may increase communications complexity for the UE. For example, if the first and second feedback messages are reported in the same uplink transmission **635**-*a*, the feedback messages may be multiplexed on a same feedback codebook (e.g., a HARQ-ACK codebook). In a first example, the PCell **605**-*a* may be unable to distinguish between the first feedback message and the second feedback message in

the uplink transmission **635**-*a*. In a second example, the UE may be configured to include additional bits distinguishing the feedback messages. In either example, reporting the first feedback message and the second feedback message in the uplink transmission **635**-*a* may reduce communications reliability and efficiency.
(91) In some examples, the scheduling configuration for cross-carrier scheduled communications on the PCell **605**-*a* may include a scheduling constraint that multiple feedback messages corresponding to downlink transmissions **630** may not be included in the same uplink transmission **635**. That is, the UE may report the first feedback message in a first uplink transmission **635** (e.g., the uplink transmission **635**-*a*) according to the scheduling configuration, and the UE may report the second feedback message in a second uplink transmission **635** different from the first uplink transmission **635** (e.g., an uplink transmission **635** different from the uplink transmission **635**-*a*) based on the scheduling constraint.
(92) FIG. **6**B illustrates an example timing diagram **600**-*b*. In the timing diagram **600**-*b*, a first DCI message in a CSS PDCCH candidate **610**-*b* on a PCell **605**-*b* may schedule a downlink transmission **630**-*c* on the PCell **605**-*b*, and a second DCI message in a USS PDCCH candidate **620**-*b* on an SCell **606**-*b* may schedule a downlink transmission **630**-*d* on the PCell **605**-*b*. The UE may report a first feedback message corresponding to the downlink transmission **630**-*c* in an uplink transmission **635**-*b*. The UE may also report a second feedback message corresponding to the downlink transmission **630**-*d* in an uplink transmission **635**-*c*. The downlink transmission **630**-*c* may be scheduled after the downlink transmission **630**-*d* and the uplink transmission **635**-*b* may be scheduled before the uplink transmission **635**-*c*. That is, the data transmissions **630**-*c* and **630**-*d* may be scheduled in a different order than the order in which the corresponding uplink transmissions **635**-*b* and **635**-*c* are scheduled, which may increase communications complexity for the UE. In a first example, the PCell **605**-*b* may fail to associate the first and second feedback messages with the data transmissions **630**-*c* and **630**-*d*, respectively, based on the different ordering. In a second example, the UE may be configured to include additional bits distinguishing the feedback messages in the uplink transmission **635**-*b* and **635**-*c* based on the different ordering. In either example, reporting the first feedback message and the second feedback message in a different order in which the corresponding data transmissions **630**-*c* and **630**-*d* are scheduled may reduce communications reliability and efficiency.
(93) In some examples, the scheduling configuration for cross-carrier scheduled communications on the PCell **605**-*b* may include a scheduling constraint that feedback messages corresponding to downlink transmissions **630** may not be reported in a different order than the order in which the downlink transmissions **630** are scheduled. That is, the downlink transmission **630**-*c* may be scheduled before the downlink transmission **630**-*d*, and the UE may be configured to report the first feedback message in an uplink transmission **635**-*b* that is scheduled before an uplink transmission **635**-*c* in which the UE is configured to report the second feedback message. Alternatively, the downlink transmission **630**-*c* may be scheduled after the downlink transmission **630**-*d*, and the UE may be configured to report the first feedback message in an uplink transmission **635**-*b* that is scheduled after an uplink transmission **635**-*c* in which the UE is configured to report the second feedback message.
(94) FIG. **7** illustrates an example of a process flow **700** that supports techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure. In some examples, the process flow **700** may implement aspects of wireless communications systems **100** and **200**. For example, the process flow **700** may include example operations associated with one or more of a PCell **705**, an SCell **706**, or a UE **715**, which may be examples of the corresponding devices described with reference to FIGS. **1** and **2**. In the following description of the process flow **700**, the operations between the PCell **705**, the SCell **706**, and the UE **715** may be performed in a different order than the example order shown, or the operations performed by the PCell **705**, the SCell **706**, and the UE **715** may be performed in different orders or

at different times. Some operations may also be omitted from the process flow **700**, and other operations may be added to the process flow **700**. The operations performed by the PCell **705**, the SCell **706**, and the UE **715** may support improvement to the UE **715** communications operations and, in some examples, may promote improvements to the UE **715** implementation of cross-carrier scheduling, among other benefits.
(95) In some examples, at **720** the UE **715** may report a monitoring capability to the SCell **706**. The monitoring capability may indicate a limit to a quantity of non-overlapping CCEs of search spaces the UE **715** is capable of monitoring, or a limit to a quantity of blind decodes of PDCCH candidates the UE **715** is capable of attempting, or both. The limit may be on a per-CC basis, where the limit may apply to the PCell **705** and the SCell **706** individually. In some examples, reporting the monitoring capability may include transmitting capability information signaling to the SCell **706**.
(96) In some examples, at **725** the SCell **706** may transmit an indication of a shared search space monitoring configuration. The indication may include an update to a previously configured shared search space monitoring configuration for the UE **715**, or the indication may configure a new shared search space monitoring configuration for the UE **715**.
(97) At **730**, the UE **715** may identify that the UE **715** is in communication with the PCell **705** and the SCell **706**. The UE **715** may communicate concurrently with the PCell **705** and an SCell **706**, for example in a CA configuration. In some examples, the UE **715** may identify the CA configuration based on the shared search space configuration indicated at **725**. Based on the shared search space monitoring configuration, the UE **715** may identify that the SCell **706** is a scheduling cell and the PCell **705** is a scheduled cell (e.g., cross-carrier scheduling). The PCell **705** and the SCell **706** may also identify the CA configuration at **730**.
(98) At **735**, the UE **715** may monitor search spaces of the PCell **705** and the SCell **706** for control information. For example, the UE **715** may monitor PDCCH candidates of one or more CSSs on the PCell **705** and PDCCH candidates of one or more USSs on the SCell **706** according to the shared search space monitoring configuration. Additionally at **735**, the PCell **705** and the SCell **706** may determine which PDCCH candidates to use for transmitting control information to the UE **715**.
(99) At **740**, the UE **715** may receive one or more DCI messages from the PCell **705** and the SCell **706**. The DCI messages may schedule the UE **715** to communicate a data transmission (e.g., a PDSCH transmission, a PUSCH transmission, etc.) with the PCell **705**. One or more DCI messages from the SCell **706** may include a cross-carrier scheduling indication that the data transmission is scheduled for the PCell **705**. In some examples, the UE **715** may receive a second DCI message from the SCell **706** scheduling a second data transmission for the SCell **706**.
(100) In some examples, at **745** the UE **715** may identify a value in a CIF of a DCI message. In some examples, the shared search space monitoring configuration may indicate which USSs of the SCell **706** the UE **715** is to monitor based on a value (e.g., 0 or 1) in a CIF of a DCI message received in a PDCCH candidate in a USS of the SCell **706**. Each USS may correspond to a value in the CIF. For example, the UE **715** may be configured to monitor a first set of USSs when the value in the CIF is 0, and a second set of USSs when the value in the CIF is 1.
(101) At **750**, the UE **715** may communicate with the PCell **705** via the scheduled data transmission in accordance with the DCI messages. In some examples, the UE **715** may communicate with the SCell **706** via the second data transmission in accordance with the **715** DCI message. The operations performed by the PCell **705**, the SCell **706**, and the UE **415** may support improvements to the UE **715** communication operations and, in some examples, may promote improvements to the UE **715** reliability, among other benefits.
(102) FIG. **8** illustrates an example of a process flow **800** that supports techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure. In some examples, the process flow **800** may implement aspects of wireless

communications systems **100** and **200**. For example, the process flow **800** may include example operations associated with one or more of a PCell **805**, an SCell **806**, or a UE **815**, which may be examples of the corresponding devices described with reference to FIGS. **1** and **2**. In the following description of the process flow **800**, the operations between the PCell **805**, the SCell **806**, and the UE **815** may be performed in a different order than the example order shown, or the operations performed by the PCell **805**, the SCell **806**, and the UE **815** may be performed in different orders or at different times. Some operations may also be omitted from the process flow **800**, and other operations may be added to the process flow **800**. The operations performed by the PCell **805**, the SCell **806**, and the UE **815** may support improvement to the UE **815** communications operations and, in some examples, may promote improvements to the UE **815** implementation of cross-carrier scheduling, among other benefits.

(103) At **820**, the UE **815** may identify that the UE **815** is in communication with the PCell **805** and the SCell **806**. The UE **815** may communicate concurrently with the PCell **805** and an SCell **806**, for example in a CA configuration. In some examples, the UE **815** may identify the CA configuration based on a shared search space configuration by the SCell **806**. Based on the shared search space monitoring configuration, the UE **815** may identify that the SCell **806** is a scheduling cell and the PCell **805** is a scheduled cell (e.g., cross-carrier scheduling). In some examples, the cross-carrier scheduled communications on the PCell **805** may be configured according to a scheduling configuration, which may include one or more scheduling constraints. The scheduling configuration and the scheduling constraints may reduce monitoring and communications complexity for the UE **815**. The UE **815** may identify the scheduling configuration, for example based on the shared search space monitoring configuration. The PCell **805** and the SCell **806** may also identify the CA configuration at **820**.

(104) At **825**, the UE **815** may monitor search spaces of the PCell **805** and the SCell **806** for control information. For example, the UE **815** may monitor PDCCH candidates of one or more CSSs on the PCell **805** and PDCCH candidates of one or more USSs on the SCell **806** according to the shared search space monitoring configuration. Additionally at **825**, the PCell **805** and the SCell **806** may determine which PDCCH candidates to use for transmitting control information to the UE **815**.

(105) At **830**, the UE **815** may receive a first DCI message based on monitoring PDCCH candidates of CSSs on the PCell **805**. The UE **815** may also receive a second DCI message based on monitoring PDCCH candidates of USSs on the SCell **806**. The DCI messages may schedule the UE **815** to communicate data transmissions with the PCell **805**. For example, the first DCI message may schedule a first data transmission and the second DCI message may schedule a second data transmission. The DCI message from the SCell **806** may include a cross-carrier scheduling indication that the data transmission is scheduled for the PCell **805**.

(106) In some examples, the scheduling configuration may include a scheduling constraint that PDCCH candidates scheduling data transmissions may not overlap in time or be in a same monitoring occasion. That is, the PCell **805** may transmit the first DCI message in a PDCCH candidate in a first duration according to the scheduling configuration, and the SCell **806** may transmit the second DCI message in a PDCCH candidate in a second duration that does not overlap in time with the first duration or is not in the same monitoring occasion as the first duration based on the scheduling constraint.

(107) In some examples, the scheduling configuration may include a scheduling constraint that PDCCH candidates scheduling data transmissions may not be in a same slot. That is, the PCell **805** may transmit the first DCI message in a PDCCH candidate in a first slot according to the scheduling configuration, and the SCell **806** may transmit the second DCI message in a PDCCH candidate in a second slot different from the first slot (e.g., in a slot before or after the first slot) based on the scheduling constraint.

(108) At **835**, the UE **815** may communicate with the PCell **805** via the first and second data

transmissions in accordance with the first and second DCI messages. In some examples, the data transmissions may include PDSCH transmissions, PUSCH transmissions, etc.
(109) In some examples, the scheduling configuration may include a scheduling constraint that the scheduled data transmissions may not overlap in time. That is, the UE **815** may communicate the first data transmission in a first duration according to the scheduling configuration, and the UE **815** may communicate the second data transmission in a second duration that does not overlap in time with the first duration based on the scheduling constraint.
(110) In some examples, the scheduling configuration include a scheduling constraint that data transmissions may not be scheduled in a same slot. That is, the UE **815** may communicate the first data transmission in a first slot according to the scheduling configuration, and the UE **815** may communicate the second data transmission in a second slot different from the first slot (a slot before or after the first slot) based on the scheduling constraint.
(111) In some examples, the scheduling configuration include a scheduling constraint that data transmissions may not be scheduled in a different order than the order in which the UE **815** receives the scheduling DCI messages. That is, the UE **815** may receive the first DCI message after receiving the second DCI message according to the scheduling configuration, and the UE **815** may be scheduled to communicate the first data transmission after communicating the second data transmission based on the scheduling constraint. Alternatively, the UE **815** may receive the first DCI message before receiving the second DCI message according to the scheduling configuration, and the UE **815** may be scheduled to communicate the first data transmission before communicating the second data transmission based on the scheduling constraint.
(112) In some examples, at **840** the UE **815** may transmit a first feedback message corresponding to the first data transmission and a second feedback message corresponding to the second data transmission. The UE **815** may transmit the feedback messages in uplink transmissions (e.g., PUSCH transmissions, PUCCH transmissions, etc.). In some examples, the feedback messages may include a HARQ-ACK message or a HARQ-NACK message.
(113) In some examples, the scheduling configuration may include a scheduling constraint that multiple feedback messages corresponding to data transmissions may not be included in a same uplink transmission. That is, the UE **815** may report the first feedback message in a first uplink transmission according to the scheduling configuration, and the UE **815** may report the second feedback message in a second uplink transmission different from the first uplink transmission based on the scheduling constraint.
(114) In some examples, the scheduling configuration may include a scheduling constraint that feedback messages corresponding to downlink transmissions may not be reported in a different order than the order in which the downlink transmissions are scheduled. That is, the first downlink transmission may be scheduled before the second downlink transmission, and the UE **815** may be configured to report the first feedback message in a first uplink transmission that is scheduled before a second uplink transmission in which the UE **815** is configured to report the second feedback message. Alternatively, the first downlink transmission may be scheduled after the second downlink transmission, and the UE **815** may be configured to report the first feedback message in a first uplink transmission that is scheduled after a second uplink transmission in which the UE **815** is configured to report the second feedback message.
(115) In some examples, at **845** the PCell **805** and the SCell **806** may, based on the feedback messages, transmit additional DCI messages scheduling additional data transmissions. The additional data transmissions may be retransmissions of the first or second data transmissions, for example based on receiving HARQ-NACK feedback.
(116) In some examples, the scheduling configuration may include a scheduling constraint that retransmissions of data transmissions are to be scheduled by the same cell (e.g., the PCell **805** or the SCell **806**) which scheduled the original data transmission. That is, the UE **815** may receive a third DCI message from the PCell **805** scheduling a retransmission of the first data transmission in

accordance with the scheduling configuration. Additionally or alternatively, the UE **815** may receive a fourth DCI message from the SCell **806** scheduling a retransmission of the second data transmission in accordance with the scheduling configuration.

(117) In some examples, at **850** the UE **815** may communicate with the PCell **805** via the scheduled retransmissions. The operations performed by the PCell **805**, the SCell **806**, and the UE **415** may support improvements to the UE **815** communication operations and, in some examples, may promote improvements to the UE **815** reliability, among other benefits.

(118) FIG. **9** shows a block diagram **900** of a device **905** that supports techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure. The device **905** may be an example of aspects of a UE as described herein. The device **905** may include a receiver **910**, a communications manager **915**, and a transmitter **920**. The device **905** may also include a processor. Each of these components may be in communication with one another (e.g., via one or more buses).

(119) The receiver **910** may receive information such as packets, user data, or control information associated with various information channels (e.g., control channels, data channels, and information related to techniques for cross-carrier scheduling from a secondary cell to a primary cell, etc.). Information may be passed on to other components of the device **905**. The receiver **910** may be an example of aspects of the transceiver **1220** described with reference to FIG. **12**. The receiver **910** may utilize a single antenna or a set of antennas.

(120) The communications manager **915** may identify that the UE is in communication with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell, monitor a common search space of the primary cell and a UE-specific search space of the secondary cell for control information in accordance with a shared search space monitoring configuration, receive one or more downlink control information messages as a result of monitoring the UE-specific search space of the secondary cell, where the one or more downlink control information messages include a cross-carrier scheduling indication that a data transmission is scheduled with the primary cell, and communicate with the primary cell via the data transmission in accordance with at least one of the one or more downlink control information messages.

(121) The communications manager **915** may also identify that the UE is in communication with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell, receive, from the primary cell, a first downlink control information message that includes a scheduling indication that a first data transmission is scheduled with the primary cell in accordance with a scheduling configuration that satisfies a set of scheduling constraints associated with the primary cell being cross-carrier scheduled by the secondary cell, receive, from the secondary cell, a second downlink control information message that includes a cross-carrier scheduling indication that a second data transmission is scheduled with the primary cell in accordance with the scheduling configuration, and communicate with the primary cell via the first data transmission in accordance with the first downlink control information message, and via the second data transmission in accordance with the second downlink control information message.

(122) The communications manager **915** as described herein may be implemented to realize one or more potential advantages. One implementation may allow the device **905** to save power and increase battery life by communicating with multiple serving cells (e.g., of one or more base stations **105** as shown in FIG. **1**) more efficiently. For example, the device **905** may efficiently communicate with the serving cells (e.g., a PCell and an SCell) in a CA configuration, as the device **905** may be able to identify search spaces to monitor based on the shared search space configuration. Additionally, the device **905** may efficiently communicate with the serving cells via cross-carrier scheduled communications configured according to a scheduling configuration, which may reduce monitoring and communications complexity for the device **905**. The communications

manager **915** may be an example of aspects of the communications manager **1210** described herein.
(123) The communications manager **915**, or its sub-components, may be implemented in hardware, code (e.g., software or firmware) executed by a processor, or any combination thereof. If implemented in code executed by a processor, the functions of the communications manager **915**, or its sub-components may be executed by a general-purpose processor, a digital signal processor (DSP), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described in the present disclosure.
(124) The communications manager **915**, or its sub-components, may be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations by one or more physical components. In some examples, the communications manager **915**, or its sub-components, may be a separate and distinct component in accordance with various aspects of the present disclosure. In some examples, the communications manager **915**, or its sub-components, may be combined with one or more other hardware components, including but not limited to an input/output (I/O) component, a transceiver, a network server, another computing device, one or more other components described in the present disclosure, or a combination thereof in accordance with various aspects of the present disclosure.
(125) The transmitter **920** may transmit signals generated by other components of the device **905**. In some examples, the transmitter **920** may be collocated with a receiver **910** in a transceiver module. For example, the transmitter **920** may be an example of aspects of the transceiver **1220** described with reference to FIG. **12**. The transmitter **920** may utilize a single antenna or a set of antennas.
(126) FIG. **10** shows a block diagram **1000** of a device **1005** that supports techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure. The device **1005** may be an example of aspects of a device **905**, or a UE as described herein. The device **1005** may include a receiver **1010**, a communications manager **1015**, and a transmitter **1045**. The device **1005** may also include a processor. Each of these components may be in communication with one another (e.g., via one or more buses).
(127) The receiver **1010** may receive information such as packets, user data, or control information associated with various information channels (e.g., control channels, data channels, and information related to techniques for cross-carrier scheduling from a secondary cell to a primary cell, etc.). Information may be passed on to other components of the device **1005**. The receiver **1010** may be an example of aspects of the transceiver **1220** described with reference to FIG. **12**. The receiver **1010** may utilize a single antenna or a set of antennas.
(128) The communications manager **1015** may be an example of aspects of the communications manager **915** as described herein. The communications manager **1015** may include a CA configuration manager **1020**, a monitoring configuration manager **1025**, a DCI manager **1030**, a data transmission manager **1035**, and a scheduling configuration manager **1040**. The communications manager **1015** may be an example of aspects of the communications manager **1210** described herein.
(129) The CA configuration manager **1020** may identify that the UE is in communication with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell.
(130) The monitoring configuration manager **1025** may monitor a common search space of the primary cell and a UE-specific search space of the secondary cell for control information in accordance with a shared search space monitoring configuration.
(131) The DCI manager **1030** may receive one or more downlink control information messages as a result of monitoring the UE-specific search space of the secondary cell, where the one or more downlink control information messages include a cross-carrier scheduling indication that a data

transmission is scheduled with the primary cell.
(132) The data transmission manager **1035** may communicate with the primary cell via the data transmission in accordance with at least one of the one or more downlink control information messages.
(133) The scheduling configuration manager **1040** may receive, from the primary cell, a first downlink control information message that includes a scheduling indication that a first data transmission is scheduled with the primary cell in accordance with a scheduling configuration that satisfies a set of scheduling constraints associated with the primary cell being cross-carrier scheduled by the secondary cell and receive, from the secondary cell, a second downlink control information message that includes a cross-carrier scheduling indication that a second data transmission is scheduled with the primary cell in accordance with the scheduling configuration.
(134) The data transmission manager **1035** may communicate with the primary cell via the first data transmission in accordance with the first downlink control information message, and via the second data transmission in accordance with the second downlink control information message.
(135) The transmitter **1045** may transmit signals generated by other components of the device **1005**. In some examples, the transmitter **1045** may be collocated with a receiver **1010** in a transceiver module. For example, the transmitter **1045** may be an example of aspects of the transceiver **1220** described with reference to FIG. **12**. The transmitter **1045** may utilize a single antenna or a set of antennas.
(136) FIG. **11** shows a block diagram **1100** of a communications manager **1105** that supports techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure. The communications manager **1105** may be an example of aspects of a communications manager **915**, a communications manager **1015**, or a communications manager **1210** described herein. The communications manager **1105** may include a CA configuration manager **1110**, a monitoring configuration manager **1115**, a DCI manager **1120**, a data transmission manager **1125**, a search space limit component **1130**, a scheduling configuration manager **1135**, and a feedback manager **1140**. Each of these modules may communicate, directly or indirectly, with one another (e.g., via one or more buses).
(137) The CA configuration manager **1110** may identify that the UE is in communication with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell.
(138) The monitoring configuration manager **1115** may monitor a common search space of the primary cell and a UE-specific search space of the secondary cell for control information in accordance with a shared search space monitoring configuration.
(139) In some examples, the monitoring configuration manager **1115** may report an indication of a capability of the UE to support shared search space control channel candidates in UE-specific search spaces not associated with the cross-carrier scheduling indication, where the shared search space monitoring configuration is based on the capability. In some examples, the monitoring configuration manager **1115** may monitor a type0 common search space of the primary cell, a type0A common search space of the primary cell, a type1 common search space of the primary cell, a type2 common search space of the primary cell, a type3 common search space of the primary cell, or any combination thereof.
(140) In some examples, the monitoring configuration manager **1115** may monitor a common search space of the primary cell and one or more UE-specific search spaces of the secondary cell for control information in accordance with a shared search space monitoring configuration, where the first downlink control information message and the second downlink control information message are received as a result of the monitoring, and where the scheduling configuration is based on the shared search space monitoring configuration.
(141) In some cases, the shared search space monitoring configuration includes control channel candidates associated with the one or more downlink control information messages that include the

cross-carrier scheduling indication. In some cases, the control channel candidates associated with the one or more downlink control information messages are not included in UE-specific search spaces not associated with the cross-carrier scheduling indication. In some cases, one or more of the control channel candidates associated with the one or more downlink control information messages are included in a second UE-specific search space not associated with the cross-carrier scheduling indication.

(142) The DCI manager **1120** may receive one or more downlink control information messages as a result of monitoring the UE-specific search space of the secondary cell, where the one or more downlink control information messages include a cross-carrier scheduling indication that a data transmission is scheduled with the primary cell. In some examples, the DCI manager **1120** may identify a respective carrier indicator field in each of the one or more downlink control information messages, where the cross-carrier scheduling indication is a value included in one of the respective carrier indicator fields. In some examples, the DCI manager **1120** may receive a second downlink control information message as a result of the monitoring, where the second downlink control information message includes a second scheduling indication that a second data transmission is scheduled with the secondary cell. In some examples, the DCI manager **1120** may receive a third downlink control information message as a result of the monitoring, where the third downlink control information message includes a third scheduling indication that a third data transmission is scheduled with the primary cell.

(143) The data transmission manager **1125** may communicate with the primary cell via the data transmission in accordance with at least one of the one or more downlink control information messages.

(144) In some examples, the data transmission manager **1125** may communicate with the primary cell via the first data transmission in accordance with the first downlink control information message, and via the second data transmission in accordance with the second downlink control information message. In some examples, the data transmission manager **1125** may communicate with the secondary cell via the second data transmission in accordance with the second downlink control information message. In some examples, the data transmission manager **1125** may communicate with the primary cell via the third data transmission in accordance with the third downlink control information message.

(145) In some examples, the data transmission manager **1125** may communicate with the primary cell via the retransmission of the first data transmission in accordance with the third downlink control information message. In some examples, the data transmission manager **1125** may communicate with the primary cell via the retransmission of the second data transmission in accordance with the fourth downlink control information message.

(146) In some examples, the data transmission manager **1125** may receive a downlink data transmission from the primary cell, transmit an uplink data transmission to the primary cell, or both.

(147) The scheduling configuration manager **1135** may receive, from the primary cell, a first downlink control information message that includes a scheduling indication that a first data transmission is scheduled with the primary cell in accordance with a scheduling configuration that satisfies a set of scheduling constraints associated with the primary cell being cross-carrier scheduled by the secondary cell. In some examples, the scheduling configuration manager **1135** may receive, from the secondary cell, a second downlink control information message that includes a cross-carrier scheduling indication that a second data transmission is scheduled with the primary cell in accordance with the scheduling configuration.

(148) In some examples, the scheduling configuration manager **1135** may receive, from the primary cell, a third downlink control information message that includes a second scheduling indication that a retransmission of the first data transmission is scheduled with the primary cell in accordance with the scheduling configuration. In some examples, the scheduling configuration manager **1135** may

receive, from the secondary cell, a fourth downlink control information message that includes a second cross-carrier scheduling indication that a retransmission of the second data transmission is scheduled with the primary cell in accordance with the scheduling configuration.

(149) In some cases, the first data transmission is communicated in a first duration in accordance with the scheduling configuration. In some cases, the second data transmission is communicated in a second duration that does not overlap with the first duration based on one or more scheduling constraints of the set of scheduling constraints.

(150) In some cases, the first data transmission is communicated in a first slot in accordance with the scheduling configuration. In some cases, the second data transmission is communicated in a second slot different from the first slot based on one or more scheduling constraints of the set of scheduling constraints.

(151) In some cases, the first downlink control information message is received before the second downlink control information message in accordance with the scheduling configuration. In some cases, the first data transmission is communicated before the second data transmission based on one or more scheduling constraints of the set of scheduling constraints.

(152) In some cases, the first downlink control information message is received after the second downlink control information message in accordance with the scheduling configuration. In some cases, the first data transmission is communicated after the second data transmission based on one or more scheduling constraints of the set of scheduling constraints.

(153) In some cases, the first downlink control information message is received in a first duration in accordance with the scheduling configuration. In some cases, the second downlink control information message is received in a second duration that does not overlap with the first duration based on one or more scheduling constraints of the set of scheduling constraints.

(154) In some cases, the first downlink control information message is received in a first monitoring occasion in accordance with the scheduling configuration. In some cases, the second downlink control information message is received in a second monitoring occasion different from the first monitoring occasion based on one or more scheduling constraints of the set of scheduling constraints.

(155) In some cases, the first downlink control information message is received in a first slot in accordance with the scheduling configuration. In some cases, the second downlink control information message is received in a second slot different from the first slot based on one or more scheduling constraints of the set of scheduling constraints.

(156) In some cases, the first feedback message is included in a first uplink transmission in accordance with the scheduling configuration. In some cases, the second feedback message is included in a second uplink transmission different from the first uplink transmission based on one or more scheduling constraints of the set of scheduling constraints.

(157) In some cases, the first data transmission is communicated before the second data transmission in accordance with the scheduling configuration. In some cases, the first feedback message is transmitted before the second feedback message based on one or more scheduling constraints of the set of scheduling constraints.

(158) In some cases, the first data transmission is communicated after the second data transmission in accordance with the scheduling configuration. In some cases, the first feedback message is transmitted after the second feedback message based on one or more scheduling constraints of the set of scheduling constraints.

(159) The search space limit component **1130** may determine a limit corresponding to the shared search space monitoring configuration, where the limit is associated with a maximum number of blind decodes to be attempted by the UE, a maximum number of non-overlapping control channel elements, or both, across control channel candidates of the common search space of the primary cell and of the UE-specific search space of the secondary cell. In some examples, the search space limit component **1130** may receive an indication of the limit from the secondary cell, where

determining the limit is based on receiving the indication.
(160) In some examples, the search space limit component **1130** may receive an indication of the shared search space monitoring configuration consistent with the limit, where the monitoring is further in accordance with the limit.
(161) In some examples, the search space limit component **1130** may receive an indication of the shared search space monitoring configuration, where the shared search space monitoring configuration includes a number of blind decodes that exceeds the maximum number of blind decodes associated with the limit, a number of non-overlapping control channel elements that exceeds the maximum number of non-overlapping control channel elements associated with the limit, or both. In some examples, the search space limit component **1130** may identify a second UE-specific search space of the secondary cell included in the shared search space monitoring configuration. In some examples, the search space limit component **1130** may refrain from monitoring the second UE-specific search space of the secondary cell in accordance with the limit.
(162) The feedback manager **1140** may transmit a first feedback message to the primary cell based on communicating with the primary cell via the first data transmission. In some examples, the feedback manager **1140** may transmit a second feedback message to the primary cell based on communicating with the primary cell via the second data transmission.
(163) In some cases, the first feedback message includes a hybrid automatic repeat request acknowledgment message or a hybrid automatic repeat request negative acknowledgment message. In some cases, the second feedback message includes a hybrid automatic repeat request acknowledgment message or a hybrid automatic repeat request negative acknowledgment message.
(164) FIG. **12** shows a diagram of a system **1200** including a device **1205** that supports techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure. The device **1205** may be an example of or include the components of device **905**, device **1005**, or a UE as described herein. The device **1205** may include components for bi-directional voice and data communications including components for transmitting and receiving communications, including a communications manager **1210**, an I/O controller **1215**, a transceiver **1220**, an antenna **1225**, memory **1230**, and a processor **1240**. These components may be in electronic communication via one or more buses (e.g., bus **1245**).
(165) The communications manager **1210** may identify that the UE is in communication with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell, monitor a common search space of the primary cell and a UE-specific search space of the secondary cell for control information in accordance with a shared search space monitoring configuration, receive one or more downlink control information messages as a result of monitoring the UE-specific search space of the secondary cell, where the one or more downlink control information messages include a cross-carrier scheduling indication that a data transmission is scheduled with the primary cell, and communicate with the primary cell via the data transmission in accordance with at least one of the one or more downlink control information messages.
(166) The communications manager **1210** may also identify that the UE is in communication with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell, receive, from the primary cell, a first downlink control information message that includes a scheduling indication that a first data transmission is scheduled with the primary cell in accordance with a scheduling configuration that satisfies a set of scheduling constraints associated with the primary cell being cross-carrier scheduled by the secondary cell, receive, from the secondary cell, a second downlink control information message that includes a cross-carrier scheduling indication that a second data transmission is scheduled with the primary cell in accordance with the scheduling configuration, and communicate with the primary cell via the first data transmission in accordance with the first downlink control information message, and via the second data transmission in accordance with the

second downlink control information message.
(167) The I/O controller **1215** may manage input and output signals for the device **1205**. The I/O controller **1215** may also manage peripherals not integrated into the device **1205**. In some cases, the I/O controller **1215** may represent a physical connection or port to an external peripheral. In some cases, the I/O controller **1215** may utilize an operating system such as iOS®, ANDROID®, MS-DOS®, MS-WINDOWS®, OS/2®, UNIX®, LINUX®, or another known operating system. In other cases, the I/O controller **1215** may represent or interact with a modem, a keyboard, a mouse, a touchscreen, or a similar device. In some cases, the I/O controller **1215** may be implemented as part of a processor. In some cases, a user may interact with the device **1205** via the I/O controller **1215** or via hardware components controlled by the I/O controller **1215**.
(168) The transceiver **1220** may communicate bi-directionally, via one or more antennas, wired, or wireless links as described above. For example, the transceiver **1220** may represent a wireless transceiver and may communicate bi-directionally with another wireless transceiver. The transceiver **1220** may also include a modem to modulate the packets and provide the modulated packets to the antennas for transmission, and to demodulate packets received from the antennas.
(169) In some cases, the wireless device may include a single antenna **1225**. However, in some cases the device may have more than one antenna **1225**, which may be capable of concurrently transmitting or receiving multiple wireless transmissions.
(170) The memory **1230** may include random-access memory (RAM) and read-only memory (ROM). The memory **1230** may store computer-readable, computer-executable code **1235** including instructions that, when executed, cause the processor to perform various functions described herein. In some cases, the memory **1230** may contain, among other things, a basic input/output system (BIOS) which may control basic hardware or software operation such as the interaction with peripheral components or devices.
(171) The processor **1240** may include an intelligent hardware device (e.g., a general-purpose processor, a DSP, a central processing unit (CPU), a microcontroller, an ASIC, an FPGA, a programmable logic device, a discrete gate or transistor logic component, a discrete hardware component, or any combination thereof). In some cases, the processor **1240** may be configured to operate a memory array using a memory controller. In other cases, a memory controller may be integrated into the processor **1240**. The processor **1240** may be configured to execute computer-readable instructions stored in a memory (e.g., the memory **1230**) to cause the device **1205** to perform various functions (e.g., functions or tasks supporting techniques for cross-carrier scheduling from a secondary cell to a primary cell).
(172) The processor **1240** of the device **1205** (e.g., controlling the receiver **910**, the transmitter **920**, or the transceiver **1220**) may reduce power consumption and increase communication efficiency based on communicating with serving cells according to the shared search space monitoring configuration and the scheduling configuration. In some examples, the processor **1240** of the device **1205** may reconfigure parameters associated with monitoring search spaces on one or more serving cells. For example, the processor **1240** of the device **1205** may turn on one or more processing units for processing the communications, increase a processing clock, or a similar mechanism within the device **1205**. As such, when scheduling messages are received, the processor **1240** may be ready to respond more efficiently through the reduction of a ramp up in processing power. The improvements in power saving and communication efficiency may further increase battery life at the device **1205** (for example, by reducing or eliminating unnecessary or failed communications, etc.).
(173) The code **1235** may include instructions to implement aspects of the present disclosure, including instructions to support wireless communications. The code **1235** may be stored in a non-transitory computer-readable medium such as system memory or other type of memory. In some cases, the code **1235** may not be directly executable by the processor **1240** but may cause a computer (e.g., when compiled and executed) to perform functions described herein.

(174) FIG. **13** shows a block diagram **1300** of a device **1305** that supports techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure. The device **1305** may be an example of aspects of a base station **105** as described herein. For example, the device **1305** may be an example of a primary cell and a secondary cell, although it is to be understood that various operations associated with the primary cell and/or the secondary cell may be performed at separate devices (e.g., different base stations **105**). Therefore, in some instances, device **1305** may be representative of a base station that includes both a primary cell and a secondary cell. In other instances, device **1305** may be representative of a base station that is acting as a primary cell, or a base station that is acting as a secondary cell. The device **1305** may include a receiver **1310**, a communications manager **1315**, and a transmitter **1320**. The device **1305** may also include a processor. Each of these components may be in communication with one another (e.g., via one or more buses).
(175) The receiver **1310** may receive information such as packets, user data, or control information associated with various information channels (e.g., control channels, data channels, and information related to techniques for cross-carrier scheduling from a secondary cell to a primary cell, etc.). Information may be passed on to other components of the device **1305**. The receiver **1310** may be an example of aspects of the transceiver **1620** described with reference to FIG. **16**. The receiver **1310** may utilize a single antenna or a set of antennas.
(176) The communications manager **1315** may configure a UE to communicate with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell, configure a common search space of the primary cell and a UE-specific search space of the secondary cell for transmitting control information in accordance with a shared search space monitoring configuration, transmit one or more downlink control information messages to the UE in the UE-specific search space of the secondary cell in accordance with the configuring, where the one or more downlink control information messages include a cross-carrier scheduling indication that a data transmission is scheduled with the primary cell, and communicate with the UE via the data transmission of the primary cell in accordance with at least one of the one or more downlink control information messages.
(177) The communications manager **1315** may also configure a UE to communicate with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell, transmit a first downlink control information message via the primary cell that includes a scheduling indication that a first data transmission is scheduled with the primary cell in accordance with a scheduling configuration that satisfies a set of scheduling constraints associated with the primary cell being cross-carrier scheduled by the secondary cell, transmit a second downlink control information message via the secondary cell that includes a cross-carrier scheduling indication that a second data transmission is scheduled with the primary cell in accordance with the scheduling configuration, and communicate with the UE via the first data transmission of the primary cell in accordance with the first downlink control information message, and via the second data transmission of the primary cell in accordance with the second downlink control information message.
(178) The communications manager **1315** as described herein may be implemented to realize one or more potential advantages. One implementation may allow the device **1305** to save power by communicating with a UE (as shown in FIG. **1**) more efficiently. For example, the device **1305** may improve reliability in communications with a UE, as the device **1305** may be able to configure shared search space monitoring and scheduling at the UE and adjust communications accordingly. The communications manager **1315** may be an example of aspects of the communications manager **1610** described herein.
(179) The communications manager **1315**, or its sub-components, may be implemented in hardware, code (e.g., software or firmware) executed by a processor, or any combination thereof. If

implemented in code executed by a processor, the functions of the communications manager **1315**, or its sub-components may be executed by a general-purpose processor, a DSP, an ASIC, an FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described in the present disclosure.

(180) The communications manager **1315**, or its sub-components, may be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations by one or more physical components. In some examples, the communications manager **1315**, or its sub-components, may be a separate and distinct component in accordance with various aspects of the present disclosure. In some examples, the communications manager **1315**, or its sub-components, may be combined with one or more other hardware components, including but not limited to an input/output (I/O) component, a transceiver, a network server, another computing device, one or more other components described in the present disclosure, or a combination thereof in accordance with various aspects of the present disclosure.

(181) The transmitter **1320** may transmit signals generated by other components of the device **1305**. In some examples, the transmitter **1320** may be collocated with a receiver **1310** in a transceiver module. For example, the transmitter **1320** may be an example of aspects of the transceiver **1620** described with reference to FIG. **16**. The transmitter **1320** may utilize a single antenna or a set of antennas.

(182) FIG. **14** shows a block diagram **1400** of a device **1405** that supports techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure. The device **1405** may be an example of aspects of a device **1305**, or a base station **105** as described herein. The device **1405** may include a receiver **1410**, a communications manager **1415**, and a transmitter **1445**. The device **1405** may also include a processor. Each of these components may be in communication with one another (e.g., via one or more buses).

(183) The receiver **1410** may receive information such as packets, user data, or control information associated with various information channels (e.g., control channels, data channels, and information related to techniques for cross-carrier scheduling from a secondary cell to a primary cell, etc.). Information may be passed on to other components of the device **1405**. The receiver **1410** may be an example of aspects of the transceiver **1620** described with reference to FIG. **16**. The receiver **1410** may utilize a single antenna or a set of antennas.

(184) The communications manager **1415** may be an example of aspects of the communications manager **1315** as described herein. The communications manager **1415** may include a CA configuration component **1420**, a search space manager **1425**, a control information manager **1430**, a data transmission component **1435**, and a scheduling manager **1440**. The communications manager **1415** may be an example of aspects of the communications manager **1610** described herein.

(185) The CA configuration component **1420** may configure a UE to communicate with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell.

(186) The search space manager **1425** may configure a common search space of the primary cell and a UE-specific search space of the secondary cell for transmitting control information in accordance with a shared search space monitoring configuration.

(187) The control information manager **1430** may transmit one or more downlink control information messages to the UE in the UE-specific search space of the secondary cell in accordance with the configuring, where the one or more downlink control information messages include a cross-carrier scheduling indication that a data transmission is scheduled with the primary cell.

(188) The data transmission component **1435** may communicate with the UE via the data

transmission of the primary cell in accordance with at least one of the one or more downlink control information messages.
(189) The CA configuration component **1420** may configure a UE to communicate with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell.
(190) The scheduling manager **1440** may transmit a first downlink control information message via the primary cell that includes a scheduling indication that a first data transmission is scheduled with the primary cell in accordance with a scheduling configuration that satisfies a set of scheduling constraints associated with the primary cell being cross-carrier scheduled by the secondary cell and transmit a second downlink control information message via the secondary cell that includes a cross-carrier scheduling indication that a second data transmission is scheduled with the primary cell in accordance with the scheduling configuration.
(191) The data transmission component **1435** may communicate with the UE via the first data transmission of the primary cell in accordance with the first downlink control information message, and via the second data transmission of the primary cell in accordance with the second downlink control information message.
(192) The transmitter **1445** may transmit signals generated by other components of the device **1405**. In some examples, the transmitter **1445** may be collocated with a receiver **1410** in a transceiver module. For example, the transmitter **1445** may be an example of aspects of the transceiver **1620** described with reference to FIG. **16**. The transmitter **1445** may utilize a single antenna or a set of antennas.
(193) FIG. **15** shows a block diagram **1500** of a communications manager **1505** that supports techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure. The communications manager **1505** may be an example of aspects of a communications manager **1315**, a communications manager **1415**, or a communications manager **1610** described herein. The communications manager **1505** may include a CA configuration component **1510**, a search space manager **1515**, a control information manager **1520**, a data transmission component **1525**, a search space limit manager **1530**, a scheduling manager **1535**, and a feedback message manager **1540**. Each of these modules may communicate, directly or indirectly, with one another (e.g., via one or more buses).
(194) The CA configuration component **1510** may configure a UE to communicate with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell.
(195) The search space manager **1515** may configure a common search space of the primary cell and a UE-specific search space of the secondary cell for transmitting control information in accordance with a shared search space monitoring configuration. In some examples, the search space manager **1515** may receive an indication of a capability of the UE to support shared search space control channel candidates in UE-specific search spaces not associated with the cross-carrier scheduling indication, where the shared search space monitoring configuration is based on receiving the indication.
(196) In some examples, the search space manager **1515** may configure a common search space of the primary cell and one or more UE-specific search spaces of the secondary cell for transmitting control information in accordance with a shared search space monitoring configuration, where the first downlink control information message and the second downlink control information message are transmitted based on the determining, and where the scheduling configuration is based on the shared search space monitoring configuration.
(197) In some cases, the shared search space monitoring configuration includes control channel candidates associated with the one or more downlink control information messages that include the cross-carrier scheduling indication. In some cases, the control channel candidates associated with the one or more downlink control information messages are not included in UE-specific search

spaces not associated with the cross-carrier scheduling indication. In some cases, one or more of the control channel candidates associated with the one or more downlink control information messages are included in a second UE-specific search space not associated with the cross-carrier scheduling indication.

(198) In some cases, the common search space of the primary cell includes a type0 common search space of the primary cell, a type0A common search space of the primary cell, a type1 common search space of the primary cell, a type2 common search space of the primary cell, a type3 common search space of the primary cell, or any combination thereof.

(199) The control information manager **1520** may transmit one or more downlink control information messages to the UE in the UE-specific search space of the secondary cell in accordance with the configuring, where the one or more downlink control information messages include a cross-carrier scheduling indication that a data transmission is scheduled with the primary cell.

(200) In some examples, the control information manager **1520** may configure a respective carrier indicator field in each of the one or more downlink control information messages, where the cross-carrier scheduling indication is a value included in one of the respective carrier indicator fields. In some examples, the control information manager **1520** may transmit a second downlink control information message, where the second downlink control information message includes a second scheduling indication that a second data transmission is scheduled with the secondary cell. In some examples, the control information manager **1520** may transmit a third downlink control information message, where the third downlink control information message includes a third scheduling indication that a third data transmission is scheduled with the primary cell.

(201) The data transmission component **1525** may communicate with the UE via the data transmission of the primary cell in accordance with at least one of the one or more downlink control information messages.

(202) In some examples, the data transmission component **1525** may communicate with the UE via the first data transmission of the primary cell in accordance with the first downlink control information message, and via the second data transmission of the primary cell in accordance with the second downlink control information message. In some examples, the data transmission component **1525** may communicate with the UE via the second data transmission of the secondary cell in accordance with the second downlink control information message. In some examples, the data transmission component **1525** may communicate with the UE via the third data transmission of the primary cell in accordance with the third downlink control information message. In some examples, the data transmission component **1525** may transmit a downlink data transmission to the UE, receive an uplink data transmission from the UE, or both.

(203) In some examples, the data transmission component **1525** may communicate with the UE via the retransmission of the first data transmission of the primary cell in accordance with the third downlink control information message. In some examples, the data transmission component **1525** may communicate with the UE via the retransmission of the second data transmission of the primary cell in accordance with the fourth downlink control information message.

(204) The scheduling manager **1535** may transmit a first downlink control information message via the primary cell that includes a scheduling indication that a first data transmission is scheduled with the primary cell in accordance with a scheduling configuration that satisfies a set of scheduling constraints associated with the primary cell being cross-carrier scheduled by the secondary cell. In some examples, the scheduling manager **1535** may transmit a second downlink control information message via the secondary cell that includes a cross-carrier scheduling indication that a second data transmission is scheduled with the primary cell in accordance with the scheduling configuration.

(205) In some examples, the scheduling manager **1535** may transmit a third downlink control information message via the primary cell that includes a second scheduling indication that a retransmission of the first data transmission is scheduled with the primary cell in accordance with

the scheduling configuration. In some examples, the scheduling manager **1535** may transmit a fourth downlink control information message via the secondary cell that includes a second cross-carrier scheduling indication that a retransmission of the second data transmission is scheduled with the primary cell in accordance with the scheduling configuration.

(206) In some cases, the first data transmission is communicated in a first duration in accordance with the scheduling configuration. In some cases, the second data transmission is communicated in a second duration that does not overlap with the first duration based on one or more scheduling constraints of the set of scheduling constraints.

(207) In some cases, the first data transmission is communicated in a first slot in accordance with the scheduling configuration. In some cases, the second data transmission is communicated in a second slot different from the first slot based on one or more scheduling constraints of the set of scheduling constraints.

(208) In some cases, the first downlink control information message is received before the second downlink control information message in accordance with the scheduling configuration. In some cases, the first data transmission is communicated before the second data transmission based on one or more scheduling constraints of the set of scheduling constraints.

(209) In some cases, the first downlink control information message is transmitted after the second downlink control information message in accordance with the scheduling configuration. In some cases, the first data transmission is communicated after the second data transmission based on one or more scheduling constraints of the set of scheduling constraints.

(210) In some cases, the first downlink control information message is transmitted in a first duration in accordance with the scheduling configuration. In some cases, the second downlink control information message is received in a second duration that does not overlap with the first duration based on one or more scheduling constraints of the set of scheduling constraints.

(211) In some cases, the first downlink control information message is transmitted in a first monitoring occasion in accordance with the scheduling configuration. In some cases, the second downlink control information message is transmitted in a second monitoring occasion different from the first monitoring occasion based on one or more scheduling constraints of the set of scheduling constraints.

(212) In some cases, the first downlink control information message is transmitted in a first slot in accordance with the scheduling configuration. In some cases, the second downlink control information message is transmitted in a second slot different from the first slot based on one or more scheduling constraints of the set of scheduling constraints.

(213) In some cases, the first feedback message is included in a first uplink transmission in accordance with the scheduling configuration. In some cases, the second feedback message is included in a second uplink transmission different from the first uplink transmission based on one or more scheduling constraints of the set of scheduling constraints.

(214) In some cases, the first data transmission is communicated before the second data transmission in accordance with the scheduling configuration. In some cases, the first feedback message is received before the second feedback message based on one or more scheduling constraints of the set of scheduling constraints.

(215) In some cases, the first data transmission is communicated after the second data transmission in accordance with the scheduling configuration. In some cases, the first feedback message is received after the second feedback message based on one or more scheduling constraints of the set of scheduling constraints.

(216) The search space limit manager **1530** may determine a limit corresponding to the shared search space monitoring configuration, where the limit is associated with a maximum number of blind decodes to be attempted by the UE, a maximum number of non-overlapping control channel elements, or both, across control channel candidates of the common search space of the primary cell and of the UE-specific search space of the secondary cell. In some examples, the search space

limit manager **1530** may transmit an indication of the limit to the UE based on determining the limit.

(217) In some examples, the search space limit manager **1530** may determine the shared search space monitoring configuration based on the limit, where the shared search space monitoring configuration is consistent with the limit. In some examples, the search space limit manager **1530** may transmit an indication of the shared search space monitoring configuration to the UE. In some examples, the search space limit manager **1530** may determine the shared search space monitoring configuration based on the limit, where the shared search space monitoring configuration includes a number of blind decodes that exceeds the maximum number of blind decodes associated with the limit, a number of non-overlapping control channel elements that exceeds the maximum number of non-overlapping control channel elements associated with the limit, or both.

(218) The feedback message manager **1540** may receive a first feedback message from the UE based on communicating with the UE via the first data transmission of the primary cell. In some examples, the feedback message manager **1540** may receive a second feedback message from the UE based on communicating with the UE via the first data transmission of the primary cell. In some cases, the first feedback message includes a hybrid automatic repeat request acknowledgment message or a hybrid automatic repeat request negative acknowledgment message. In some cases, the second feedback message includes a hybrid automatic repeat request acknowledgment message or a hybrid automatic repeat request negative acknowledgment message.

(219) FIG. **16** shows a diagram of a system **1600** including a device **1605** that supports techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure. The device **1605** may be an example of or include the components of device **1305**, device **1405**, or a base station **105** as described herein. The device **1605** may include components for bi-directional voice and data communications including components for transmitting and receiving communications, including a communications manager **1610**, a network communications manager **1615**, a transceiver **1620**, an antenna **1625**, memory **1630**, a processor **1640**, and an inter-station communications manager **1645**. These components may be in electronic communication via one or more buses (e.g., bus **1650**).

(220) The communications manager **1610** may configure a UE to communicate with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell, configure a common search space of the primary cell and a UE-specific search space of the secondary cell for transmitting control information in accordance with a shared search space monitoring configuration, transmit one or more downlink control information messages to the UE in the UE-specific search space of the secondary cell in accordance with the configuring, where the one or more downlink control information messages include a cross-carrier scheduling indication that a data transmission is scheduled with the primary cell, and communicate with the UE via the data transmission of the primary cell in accordance with at least one of the one or more downlink control information messages.

(221) The communications manager **1610** may also configure a UE to communicate with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell, transmit a first downlink control information message via the primary cell that includes a scheduling indication that a first data transmission is scheduled with the primary cell in accordance with a scheduling configuration that satisfies a set of scheduling constraints associated with the primary cell being cross-carrier scheduled by the secondary cell, transmit a second downlink control information message via the secondary cell that includes a cross-carrier scheduling indication that a second data transmission is scheduled with the primary cell in accordance with the scheduling configuration, and communicate with the UE via the first data transmission of the primary cell in accordance with the first downlink control information message, and via the second data transmission of the primary

cell in accordance with the second downlink control information message.
(222) The network communications manager **1615** may manage communications with the core network (e.g., via one or more wired backhaul links). For example, the network communications manager **1615** may manage the transfer of data communications for client devices, such as one or more UEs **115**.
(223) The transceiver **1620** may communicate bi-directionally, via one or more antennas, wired, or wireless links as described above. For example, the transceiver **1620** may represent a wireless transceiver and may communicate bi-directionally with another wireless transceiver. The transceiver **1620** may also include a modem to modulate the packets and provide the modulated packets to the antennas for transmission, and to demodulate packets received from the antennas.
(224) In some cases, the wireless device may include a single antenna **1625**. However, in some cases the device may have more than one antenna **1625**, which may be capable of concurrently transmitting or receiving multiple wireless transmissions.
(225) The memory **1630** may include RAM, ROM, or a combination thereof. The memory **1630** may store computer-readable code **1635** including instructions that, when executed by a processor (e.g., the processor **1640**) cause the device to perform various functions described herein. In some cases, the memory **1630** may contain, among other things, a BIOS which may control basic hardware or software operation such as the interaction with peripheral components or devices.
(226) The processor **1640** may include an intelligent hardware device (e.g., a general-purpose processor, a DSP, a CPU, a microcontroller, an ASIC, an FPGA, a programmable logic device, a discrete gate or transistor logic component, a discrete hardware component, or any combination thereof). In some cases, the processor **1640** may be configured to operate a memory array using a memory controller. In some cases, a memory controller may be integrated into processor **1640**. The processor **1640** may be configured to execute computer-readable instructions stored in a memory (e.g., the memory **1630**) to cause the device **1605** to perform various functions (e.g., functions or tasks supporting techniques for cross-carrier scheduling from a secondary cell to a primary cell).
(227) The inter-station communications manager **1645** may manage communications with other base station **105**, and may include a controller or scheduler for controlling communications with UEs **115** in cooperation with other base stations **105**. For example, the inter-station communications manager **1645** may coordinate scheduling for transmissions to UEs **115** for various interference mitigation techniques such as beamforming or joint transmission. In some examples, the inter-station communications manager **1645** may provide an X2 interface within an LTE/LTE-A wireless communication network technology to provide communication between base stations **105**.
(228) The code **1635** may include instructions to implement aspects of the present disclosure, including instructions to support wireless communications. The code **1635** may be stored in a non-transitory computer-readable medium such as system memory or other type of memory. In some cases, the code **1635** may not be directly executable by the processor **1640** but may cause a computer (e.g., when compiled and executed) to perform functions described herein.
(229) FIG. **17** shows a flowchart illustrating a method **1700** that supports techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure. The operations of method **1700** may be implemented by a UE or its components as described herein. For example, the operations of method **1700** may be performed by a communications manager as described with reference to FIGS. **9** through **12**. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the functions described below. Additionally or alternatively, a UE may perform aspects of the functions described below using special-purpose hardware.
(230) At **1705**, the UE may identify that the UE is in communication with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell. The operations of **1705** may

be performed according to the methods described herein. In some examples, aspects of the operations of **1705** may be performed by a CA configuration manager as described with reference to FIGS. **9** through **12**.
(231) At **1710**, the UE may monitor a common search space of the primary cell and a UE-specific search space of the secondary cell for control information in accordance with a shared search space monitoring configuration. The operations of **1710** may be performed according to the methods described herein. In some examples, aspects of the operations of **1710** may be performed by a monitoring configuration manager as described with reference to FIGS. **9** through **12**.
(232) At **1715**, the UE may receive one or more downlink control information messages as a result of monitoring the UE-specific search space of the secondary cell, where the one or more downlink control information messages include a cross-carrier scheduling indication that a data transmission is scheduled with the primary cell. The operations of **1715** may be performed according to the methods described herein. In some examples, aspects of the operations of **1715** may be performed by a DCI manager as described with reference to FIGS. **9** through **12**.
(233) At **1720**, the UE may communicate with the primary cell via the data transmission in accordance with at least one of the one or more downlink control information messages. The operations of **1720** may be performed according to the methods described herein. In some examples, aspects of the operations of **1720** may be performed by a data transmission manager as described with reference to FIGS. **9** through **12**.
(234) FIG. **18** shows a flowchart illustrating a method **1800** that supports techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure. The operations of method **1800** may be implemented by a base station **105** or its components as described herein. For example, the operations of method **1800** may be performed by a communications manager as described with reference to FIGS. **13** through **16**. In some examples, a base station may execute a set of instructions to control the functional elements of the base station to perform the functions described below. Additionally or alternatively, a base station may perform aspects of the functions described below using special-purpose hardware.
(235) At **1805**, the base station may configure a UE to communicate with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell. The operations of **1805** may be performed according to the methods described herein. In some examples, aspects of the operations of **1805** may be performed by a CA configuration component as described with reference to FIGS. **13** through **16**.
(236) At **1810**, the base station may configure a common search space of the primary cell and a UE-specific search space of the secondary cell for transmitting control information in accordance with a shared search space monitoring configuration. The operations of **1810** may be performed according to the methods described herein. In some examples, aspects of the operations of **1810** may be performed by a search space manager as described with reference to FIGS. **13** through **16**.
(237) At **1815**, the base station may transmit one or more downlink control information messages to the UE in the UE-specific search space of the secondary cell in accordance with the configuring, where the one or more downlink control information messages include a cross-carrier scheduling indication that a data transmission is scheduled with the primary cell. The operations of **1815** may be performed according to the methods described herein. In some examples, aspects of the operations of **1815** may be performed by a control information manager as described with reference to FIGS. **13** through **16**.
(238) At **1820**, the base station may communicate with the UE via the data transmission of the primary cell in accordance with at least one of the one or more downlink control information messages. The operations of **1820** may be performed according to the methods described herein. In some examples, aspects of the operations of **1820** may be performed by a data transmission component as described with reference to FIGS. **13** through **16**.

(239) FIG. **19** shows a flowchart illustrating a method **1900** that supports techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure. The operations of method **1900** may be implemented by a UE or its components as described herein. For example, the operations of method **1900** may be performed by a communications manager as described with reference to FIGS. **9** through **12**. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the functions described below. Additionally or alternatively, a UE may perform aspects of the functions described below using special-purpose hardware.
(240) At **1905**, the UE may identify that the UE is in communication with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell. The operations of **1905** may be performed according to the methods described herein. In some examples, aspects of the operations of **1905** may be performed by a CA configuration manager as described with reference to FIGS. **9** through **12**.
(241) At **1910**, the UE may receive, from the primary cell, a first downlink control information message that includes a scheduling indication that a first data transmission is scheduled with the primary cell in accordance with a scheduling configuration that satisfies a set of scheduling constraints associated with the primary cell being cross-carrier scheduled by the secondary cell. The operations of **1910** may be performed according to the methods described herein. In some examples, aspects of the operations of **1910** may be performed by a scheduling configuration manager as described with reference to FIGS. **9** through **12**.
(242) At **1915**, the UE may receive, from the secondary cell, a second downlink control information message that includes a cross-carrier scheduling indication that a second data transmission is scheduled with the primary cell in accordance with the scheduling configuration. The operations of **1915** may be performed according to the methods described herein. In some examples, aspects of the operations of **1915** may be performed by a scheduling configuration manager as described with reference to FIGS. **9** through **12**.
(243) At **1920**, the UE may communicate with the primary cell via the first data transmission in accordance with the first downlink control information message, and via the second data transmission in accordance with the second downlink control information message. The operations of **1920** may be performed according to the methods described herein. In some examples, aspects of the operations of **1920** may be performed by a data transmission manager as described with reference to FIGS. **9** through **12**.
(244) FIG. **20** shows a flowchart illustrating a method **2000** that supports techniques for cross-carrier scheduling from a secondary cell to a primary cell in accordance with aspects of the present disclosure. The operations of method **2000** may be implemented by a base station **105** or its components as described herein. For example, the operations of method **2000** may be performed by a communications manager as described with reference to FIGS. **13** through **16**. In some examples, a base station may execute a set of instructions to control the functional elements of the base station to perform the functions described below. Additionally or alternatively, a base station may perform aspects of the functions described below using special-purpose hardware.
(245) At **2005**, the base station may configure a UE to communicate with a primary cell and a secondary cell in a carrier aggregation configuration, where the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell. The operations of **2005** may be performed according to the methods described herein. In some examples, aspects of the operations of **2005** may be performed by a CA configuration component as described with reference to FIGS. **13** through **16**.
(246) At **2010**, the base station may transmit a first downlink control information message via the primary cell that includes a scheduling indication that a first data transmission is scheduled with the primary cell in accordance with a scheduling configuration that satisfies a set of scheduling

constraints associated with the primary cell being cross-carrier scheduled by the secondary cell. The operations of **2010** may be performed according to the methods described herein. In some examples, aspects of the operations of **2010** may be performed by a scheduling manager as described with reference to FIGS. **13** through **16**.

(247) At **2015**, the base station may transmit a second downlink control information message via the secondary cell that includes a cross-carrier scheduling indication that a second data transmission is scheduled with the primary cell in accordance with the scheduling configuration. The operations of **2015** may be performed according to the methods described herein. In some examples, aspects of the operations of **2015** may be performed by a scheduling manager as described with reference to FIGS. **13** through **16**.

(248) At **2020**, the base station may communicate with the UE via the first data transmission of the primary cell in accordance with the first downlink control information message, and via the second data transmission of the primary cell in accordance with the second downlink control information message. The operations of **2020** may be performed according to the methods described herein. In some examples, aspects of the operations of **2020** may be performed by a data transmission component as described with reference to FIGS. **13** through **16**.

(249) It should be noted that the methods described herein describe possible implementations, and that the operations and the steps may be rearranged or otherwise modified and that other implementations are possible. Further, aspects from two or more of the methods may be combined.

(250) The following provides an overview of aspects of the present disclosure: Aspect 1: A method for wireless communications at a UE, comprising: identifying that the UE is in communication with a primary cell and a secondary cell in a carrier aggregation configuration, wherein the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell; monitoring a common search space of the primary cell and a UE-specific search space of the secondary cell for control information in accordance with a shared search space monitoring configuration; receiving one or more downlink control information messages as a result of monitoring the UE-specific search space of the secondary cell, wherein the one or more downlink control information messages include a cross-carrier scheduling indication that a data transmission is scheduled with the primary cell; and communicating with the primary cell via the data transmission in accordance with at least one of the one or more downlink control information messages. Aspect 2: The method of aspect 1, further comprising: identifying a respective carrier indicator field in each of the one or more downlink control information messages, wherein the cross-carrier scheduling indication is a value included in one of the respective carrier indicator fields. Aspect 3: The method of any of aspects 1 or 2, further comprising: determining a limit corresponding to the shared search space monitoring configuration, wherein the limit is associated with a maximum number of blind decodes to be attempted by the UE, a maximum number of non-overlapping control channel elements, or both, across control channel candidates of the common search space of the primary cell and of the UE-specific search space of the secondary cell. Aspect 4: The method of any of aspects 1 to 3, further comprising: receiving an indication of the limit from the secondary cell, wherein determining the limit is based at least in part on receiving the indication. Aspect 5: The method of any of aspects 1 to 4, further comprising: receiving an indication of the shared search space monitoring configuration consistent with the limit, wherein the monitoring is further in accordance with the limit. Aspect 6: The method of any of aspects 1 to 4, further comprising: receiving an indication of the shared search space monitoring configuration, wherein the shared search space monitoring configuration includes a number of blind decodes that exceeds the maximum number of blind decodes associated with the limit, a number of non-overlapping control channel elements that exceeds the maximum number of non-overlapping control channel elements associated with the limit, or both; identifying a second UE-specific search space of the secondary cell included in the shared search space monitoring configuration; and refraining from monitoring the second UE-specific search space of the secondary cell in

accordance with the limit. Aspect 7: The method of any of aspects 1 to 6, further comprising: receiving a second downlink control information message as a result of the monitoring, wherein the second downlink control information message includes a second scheduling indication that a second data transmission is scheduled with the secondary cell; and communicating with the secondary cell via the second data transmission in accordance with the second downlink control information message. Aspect 8: The method of any of aspects 1 to 7, further comprising: receiving a third downlink control information message as a result of the monitoring, wherein the third downlink control information message includes a third scheduling indication that a third data transmission is scheduled with the primary cell; and communicating with the primary cell via the third data transmission in accordance with the third downlink control information message. Aspect 9: The method of any of aspects 1 to 8, wherein: the shared search space monitoring configuration includes control channel candidates associated with the one or more downlink control information messages that include the cross-carrier scheduling indication; and the control channel candidates associated with the one or more downlink control information messages are not included in UE-specific search spaces not associated with the cross-carrier scheduling indication. Aspect 10: The method of any of aspects 1 to 8, wherein: the shared search space monitoring configuration includes control channel candidates associated with the one or more downlink control information messages that include the cross-carrier scheduling indication; and one or more of the control channel candidates associated with the one or more downlink control information messages are included in a second UE-specific search space not associated with the cross-carrier scheduling indication. Aspect 11: The method of any of aspects 1 to 10, further comprising: reporting an indication of a capability of the UE to support shared search space control channel candidates in UE-specific search spaces not associated with the cross-carrier scheduling indication, wherein the shared search space monitoring configuration is based at least in part on the capability. Aspect 12: The method of any of aspects 1 to 11, wherein monitoring the common search space of the primary cell for the control information comprises: monitoring a type0 common search space of the primary cell, a type0A common search space of the primary cell, a type1 common search space of the primary cell, a type2 common search space of the primary cell, a type3 common search space of the primary cell, or any combination thereof. Aspect 13: The method of any of aspects 1 to 12, wherein communicating with the primary cell via the data transmission comprises: receiving a downlink data transmission from the primary cell, transmitting an uplink data transmission to the primary cell, or both. Aspect 14: An apparatus for wireless communications comprising a processor; memory in electronic communication with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform a method of any of aspects 1 to 13. Aspect 15: A non-transitory computer-readable medium storing code for wireless communications, the code comprising instructions executable by a processor to perform a method of any of aspects 1 to 13. Aspect 16: An apparatus, comprising means for performing the method of any of aspects 1 to 13. Aspect 17: A method for wireless communications at a base station, comprising: configuring a UE to communicate with a primary cell and a secondary cell in a carrier aggregation configuration, wherein the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell; configuring a common search space of the primary cell and a UE-specific search space of the secondary cell for transmitting control information in accordance with a shared search space monitoring configuration; transmitting one or more downlink control information messages to the UE in the UE-specific search space of the secondary cell in accordance with the configuring, wherein the one or more downlink control information messages include a cross-carrier scheduling indication that a data transmission is scheduled with the primary cell; and communicating with the UE via the data transmission of the primary cell in accordance with at least one of the one or more downlink control information messages. Aspect 18: The method of aspect 17, further comprising: configuring a respective carrier indicator field in each of the one or more downlink control information messages, wherein the cross-carrier scheduling indication is a value included in one of the respective carrier indicator fields. Aspect 19: The method of any of aspects 17 or 18, further comprising: determining a limit corresponding to the shared search space monitoring configuration, wherein the limit is associated with a maximum number of blind decodes to be attempted by the UE, a maximum number of non-overlapping control channel elements, or both, across control channel candidates of the common search space of the primary cell and of the UE-specific search space of the secondary cell. Aspect 20: The method of any of aspects 17 to 19, further comprising: transmitting an indication of the limit to the UE based at least in part on determining the limit. Aspect 21: The method of any of aspects 17 to 20, further comprising: determining the shared search space monitoring configuration based at least in part on the limit, wherein the shared search space monitoring configuration is consistent with the limit; and transmitting an indication of the shared search space monitoring configuration to the UE. Aspect 22: The method of any of aspects 17 to 20, further comprising: determining the shared search space monitoring configuration based at least in part on the limit, wherein the shared search space monitoring configuration includes a number of blind decodes that exceeds the maximum number of blind decodes associated with the limit, a number of non-overlapping control channel elements that exceeds the maximum number of non-overlapping control channel elements associated with the limit, or both; and transmitting an indication of the shared search space monitoring configuration to the UE. Aspect 23: The method of any of aspects 17 to 22, further comprising: transmitting a second downlink control information message, wherein the second downlink control information message includes a second scheduling indication that a second data transmission is scheduled with the secondary cell; and communicating with the UE via the second data transmission of the secondary cell in accordance with the second downlink control information message. Aspect 24: The method of any of aspects 17 to 23, further comprising: transmitting a third downlink control information message, wherein the third downlink control information message includes a third scheduling indication that a third data transmission is scheduled with the primary cell; and communicating with the UE via the third data transmission of the primary cell in accordance with the third downlink control information message. Aspect 25: The method of any of aspects 17 to 24, wherein: the shared search space monitoring configuration includes control channel candidates associated with the one or more downlink control information messages that include the cross-carrier scheduling indication; and the control channel candidates associated with the one or more downlink control information messages are not included in UE-specific search spaces not associated with the cross-carrier scheduling indication. Aspect 26: The method of any of aspects 17 to 24, wherein: the shared search space monitoring configuration includes control channel candidates associated with the one or more downlink control information messages that include the cross-carrier scheduling indication; and one or more of the control channel candidates associated with the one or more downlink control information messages are included in a second UE-specific search space not associated with the cross-carrier scheduling indication. Aspect 27: The method of any of aspects 17 to 26, further comprising: receiving an indication of a capability of the UE to support shared search space control channel candidates in UE-specific search spaces not associated with the cross-carrier scheduling indication, wherein the shared search space monitoring configuration is based at least in part on receiving the indication. Aspect 28: The method of any of aspects 17 to 27, wherein the common search space of the primary cell comprises a type0 common search space of the primary cell, a type0A common search space of the primary cell, a type1 common search space of the primary cell, a type2 common search space of the primary cell, a type3 common search space of the primary cell, or any combination thereof. Aspect 29: The method of any of aspects 17 to 28, wherein communicating with the UE via the data transmission of the primary cell comprises: transmitting a downlink data transmission to the UE, receiving an uplink data transmission from the UE, or both. Aspect 30: An apparatus for wireless communications comprising a processor; memory in electronic communication with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform a method of any of aspects 17 to 29. Aspect 31: A non-transitory computer-readable medium storing code for wireless communications, the code comprising instructions executable by a processor to perform a method of any of aspects 17 to 29. Aspect 32: An apparatus, comprising means for performing the method of any of aspects 17 to 29. Aspect 33: A method for wireless communications at a UE, comprising: identifying that the UE is in communication with a primary cell and a secondary cell in a carrier aggregation configuration, wherein the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell; receiving, from the primary cell, a first downlink control information message that includes a scheduling indication that a first data transmission is scheduled with the primary cell in accordance with a scheduling configuration that satisfies a plurality of scheduling constraints associated with the primary cell being cross-carrier scheduled by the secondary cell; receiving, from the secondary cell, a second downlink control information message that includes a cross-carrier scheduling indication that a second data transmission is scheduled with the primary cell in accordance with the scheduling configuration; and communicating with the primary cell via the first data transmission in accordance with the first downlink control information message, and via the second data transmission in accordance with the second downlink control information message. Aspect 34: The method of aspect 33, further comprising: monitoring a common search space of the primary cell and one or more UE-specific search spaces of the secondary cell for control information in accordance with a shared search space monitoring configuration, wherein the first downlink control information message and the second downlink control information message are received as a result of the monitoring, and wherein the scheduling configuration is based at least in part on the shared search space monitoring configuration. Aspect 35: The method of any of aspects 33 or 34, wherein: the first data transmission is communicated in a first duration in accordance with the scheduling configuration; and the second data transmission is communicated in a second duration that does not overlap with the first duration based at least in part on one or more scheduling constraints of the plurality of scheduling constraints. Aspect 36: The method of any of aspects 33 to 35, wherein: the first data transmission is communicated in a first slot in accordance with the scheduling configuration; and the second data transmission is communicated in a second slot different from the first slot based at least in part on one or more scheduling constraints of the plurality of scheduling constraints. Aspect 37: The method of any of aspects 33 to 36, wherein: the first downlink control information message is received before the second downlink control information message in accordance with the scheduling configuration; and the first data transmission is communicated before the second data transmission based at least in part on one or more scheduling constraints of the plurality of scheduling constraints. Aspect 38: The method of any of aspects 33 to 36, wherein: the first downlink control information message is received after the second downlink control information message in accordance with the scheduling configuration; and the first data transmission is communicated after the second data transmission based at least in part on one or more scheduling constraints of the plurality of scheduling constraints. Aspect 39: The method of any of aspects 33 to 38, wherein: the first downlink control information message is received in a first duration in accordance with the scheduling configuration; and the second downlink control information message is received in a second duration that does not overlap with the first duration based at least in part on one or more scheduling constraints of the plurality of scheduling constraints. Aspect 40: The method of any of aspects 33 to 39, wherein: the first downlink control information message is received in a first monitoring occasion in accordance with the scheduling configuration; and the second downlink control information message is received in a second monitoring occasion different from the first monitoring occasion based at least in part on one or more scheduling constraints of the plurality of scheduling constraints. Aspect 41: The method of any of aspects 33 to 40, wherein: the first downlink control information message is received in a first slot in accordance with the scheduling configuration; and the second downlink control information message is received in a second slot different from the first slot based at least in part on one or more scheduling constraints of the plurality of scheduling constraints. Aspect 42: The method of any of aspects 33 to 41, further comprising: transmitting a first feedback message to the primary cell based at least in part on communicating with the primary cell via the first data transmission; and transmitting a second feedback message to the primary cell based at least in part on communicating with the primary cell via the second data transmission. Aspect 43: The method of any of aspects 33 to 42, wherein: the first feedback message is included in a first uplink transmission in accordance with the scheduling configuration; and the second feedback message is included in a second uplink transmission different from the first uplink transmission based at least in part on one or more scheduling constraints of the plurality of scheduling constraints. Aspect 44: The method of any of aspects 33 to 43, wherein: the first data transmission is communicated before the second data transmission in accordance with the scheduling configuration; and the first feedback message is transmitted before the second feedback message based at least in part on one or more scheduling constraints of the plurality of scheduling constraints. Aspect 45: The method of any of aspects 33 to 43, wherein: the first data transmission is communicated after the second data transmission in accordance with the scheduling configuration; and the first feedback message is transmitted after the second feedback message based at least in part on one or more scheduling constraints of the plurality of scheduling constraints. Aspect 46: The method of any of aspects 33 to 45, further comprising: receiving, from the primary cell, a third downlink control information message that includes a second scheduling indication that a retransmission of the first data transmission is scheduled with the primary cell in accordance with the scheduling configuration; and communicating with the primary cell via the retransmission of the first data transmission in accordance with the third downlink control information message. Aspect 47: The method of any of aspects 33 to 46, further comprising: receiving, from the secondary cell, a fourth downlink control information message that includes a second cross-carrier scheduling indication that a retransmission of the second data transmission is scheduled with the primary cell in accordance with the scheduling configuration; and communicating with the primary cell via the retransmission of the second data transmission in accordance with the fourth downlink control information message. Aspect 48: The method of any of aspects 33 to 47, wherein: the first feedback message comprises a hybrid automatic repeat request acknowledgment message or a hybrid automatic repeat request negative acknowledgment message; and the second feedback message comprises a hybrid automatic repeat request acknowledgment message or a hybrid automatic repeat request negative acknowledgment message. Aspect 49: The method of any of aspects 33 to 48, wherein communicating with the primary cell via a data transmission comprises: receiving a downlink data transmission from the primary cell, transmitting an uplink data transmission to the primary cell, or both. Aspect 50: An apparatus for wireless communications comprising a processor; memory in electronic communication with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform a method of any of aspects 33 to 49. Aspect 51: A non-transitory computer-readable medium storing code for wireless communications, the code comprising instructions executable by a processor to perform a method of any of aspects 33 to 49. Aspect 52: An apparatus, comprising means for performing the method of any of aspects 33 to 49. Aspect 53: A method for wireless communications at a base station, comprising: configuring a UE to communicate with a primary cell and a secondary cell in a carrier aggregation configuration, wherein the primary cell is a cross-carrier scheduled cell and the secondary cell is a cross-carrier scheduling cell; transmitting a first downlink control information message via the primary cell that includes a scheduling indication that a first data transmission is scheduled with the primary cell in accordance with a scheduling configuration that satisfies a plurality of scheduling constraints associated with the primary cell being cross-carrier scheduled by the secondary cell; transmitting a second downlink control information message via the secondary cell that includes a cross-carrier scheduling indication that a second data transmission is scheduled with the primary cell in accordance with the scheduling configuration; and communicating with the UE via the first data transmission of the primary cell in accordance with the first downlink control information message, and via the second data transmission of the primary cell in accordance with the second downlink control information message. Aspect 54: The method of aspect 53, further comprising: configuring a common search space of the primary cell and one or more UE-specific search spaces of the secondary cell for transmitting control information in accordance with a shared search space monitoring configuration, wherein the first downlink control information message and the second downlink control information message are transmitted based at least in part on the determining, and wherein the scheduling configuration is based at least in part on the shared search space monitoring configuration. Aspect 55: The method of any of aspects 53 or 54, wherein: the first data transmission is communicated in a first duration in accordance with the scheduling configuration; and the second data transmission is communicated in a second duration that does not overlap with the first duration based at least in part on one or more scheduling constraints of the plurality of scheduling constraints. Aspect 56: The method of any of aspects 53 to 55, wherein: the first data transmission is communicated in a first slot in accordance with the scheduling configuration; and the second data transmission is communicated in a second slot different from the first slot based at least in part on one or more scheduling constraints of the plurality of scheduling constraints. Aspect 57: The method of any of aspects 53 to 56, wherein: the first downlink control information message is received before the second downlink control information message in accordance with the scheduling configuration; and the first data transmission is communicated before the second data transmission based at least in part on one or more scheduling constraints of the plurality of scheduling constraints. Aspect 58: The method of any of aspects 53 to 56, wherein: the first downlink control information message is transmitted after the second downlink control information message in accordance with the scheduling configuration; and the first data transmission is communicated after the second data transmission based at least in part on one or more scheduling constraints of the plurality of scheduling constraints. Aspect 59: The method of any of aspects 53 to 58, wherein: the first downlink control information message is transmitted in a first duration in accordance with the scheduling configuration; and the second downlink control information message is received in a second duration that does not overlap with the first duration based at least in part on one or more scheduling constraints of the plurality of scheduling constraints. Aspect 60: The method of any of aspects 53 to 59, wherein: the first downlink control information message is transmitted in a first monitoring occasion in accordance with the scheduling configuration; and the second downlink control information message is transmitted in a second monitoring occasion different from the first monitoring occasion based at least in part on one or more scheduling constraints of the plurality of scheduling constraints. Aspect 61: The method of any of aspects 53 to 60, wherein: the first downlink control information message is transmitted in a first slot in accordance with the scheduling configuration; and the second downlink control information message is transmitted in a second slot different from the first slot based at least in part on one or more scheduling constraints of the plurality of scheduling constraints. Aspect 62: The method of any of aspects 53 to 61, further comprising: receiving a first feedback message from the UE based at least in part on communicating with the UE via the first data transmission of the primary cell; and receiving a second feedback message from the UE based at least in part on communicating with the UE via the first data transmission of the primary cell. Aspect 63: The method of any of aspects 53 to 62, wherein: the first feedback message is included in a first uplink transmission in accordance with the scheduling configuration; and the second feedback message is included in a second uplink transmission different from the first uplink transmission based at least in part on one or more scheduling constraints of the plurality of scheduling constraints. Aspect 64: The method of any of aspects 53 to 63, wherein: the first data transmission is communicated before the second data transmission in accordance with the scheduling configuration; and the first feedback message is received before the second feedback message based at least in part on one or more scheduling constraints of the plurality of scheduling constraints. Aspect 65: The method of any of aspects 53 to

63, wherein: the first data transmission is communicated after the second data transmission in accordance with the scheduling configuration; and the first feedback message is received after the second feedback message based at least in part on one or more scheduling constraints of the plurality of scheduling constraints. Aspect 66: The method of any of aspects 53 to 65, further comprising: transmitting a third downlink control information message via the primary cell that includes a second scheduling indication that a retransmission of the first data transmission is scheduled with the primary cell in accordance with the scheduling configuration; and communicating with the UE via the retransmission of the first data transmission of the primary cell in accordance with the third downlink control information message. Aspect 67: The method of any of aspects 53 to 66, further comprising: transmitting a fourth downlink control information message via the secondary cell that includes a second cross-carrier scheduling indication that a retransmission of the second data transmission is scheduled with the primary cell in accordance with the scheduling configuration; and communicating with the UE via the retransmission of the second data transmission of the primary cell in accordance with the fourth downlink control information message. Aspect 68: The method of any of aspects 53 to 67, wherein: the first feedback message comprises a hybrid automatic repeat request acknowledgment message or a hybrid automatic repeat request negative acknowledgment message; and the second feedback message comprises a hybrid automatic repeat request acknowledgment message or a hybrid automatic repeat request negative acknowledgment message. Aspect 69: The method of any of aspects 53 to 68, wherein communicating with the UE via a data transmission of the primary cell comprises: transmitting a downlink data transmission to the UE, receiving an uplink data transmission from the UE, or both. Aspect 70: An apparatus for wireless communications comprising a processor; memory in electronic communication with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform a method of any of aspects 53 to 69. Aspect 71: A non-transitory computer-readable medium storing code for wireless communications, the code comprising instructions executable by a processor to perform a method of any of aspects 53 to 69. Aspect 72: An apparatus, comprising means for performing the method of any of aspects 53 to 69.

(251) Although aspects of an LTE, LTE-A, LTE-A Pro, or NR system may be described for purposes of example, and LTE, LTE-A, LTE-A Pro, or NR terminology may be used in much of the description, the techniques described herein are applicable beyond LTE, LTE-A, LTE-A Pro, or NR networks. For example, the described techniques may be applicable to various other wireless communications systems such as Ultra Mobile Broadband (UMB), Institute of Electrical and Electronics Engineers (IEEE) 802.11 (Wi-Fi), IEEE 802.16 (WiMAX), IEEE 802.20, Flash-OFDM, as well as other systems and radio technologies not explicitly mentioned herein.

(252) Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

(253) The various illustrative blocks and components described in connection with the disclosure herein may be implemented or performed with a general-purpose processor, a DSP, an ASIC, a CPU, an FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

(254) The functions described herein may be implemented in hardware, software executed by a

processor, firmware, or any combination thereof. If implemented in software executed by a processor, the functions may be stored on or transmitted over as one or more instructions or code on a computer-readable medium. Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described herein may be implemented using software executed by a processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations.

(255) Computer-readable media includes both non-transitory computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A non-transitory storage medium may be any available medium that may be accessed by a general-purpose or special-purpose computer. By way of example, and not limitation, non-transitory computer-readable media may include RAM, ROM, electrically erasable programmable ROM (EEPROM), flash memory, compact disk (CD) ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other non-transitory medium that may be used to carry or store desired program code means in the form of instructions or data structures and that may be accessed by a general-purpose or special-purpose computer, or a general-purpose or special-purpose processor. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of computer-readable medium. Disk and disc, as used herein, include CD, laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above are also included within the scope of computer-readable media.

(256) As used herein, including in the claims, “or” as used in a list of items (e.g., a list of items prefaced by a phrase such as “at least one of” or “one or more of”) indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase “based on” shall not be construed as a reference to a closed set of conditions. For example, an example step that is described as “based on condition A” may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase “based on” shall be construed in the same manner as the phrase “based at least in part on.”

(257) In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If just the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label, or other subsequent reference label.

(258) The description set forth herein, in connection with the appended drawings, describes example configurations and does not represent all the examples that may be implemented or that are within the scope of the claims. The term “example” used herein means “serving as an example, instance, or illustration,” and not “preferred” or “advantageous over other examples.” The detailed description includes specific details for the purpose of providing an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some instances, known structures and devices are shown in block diagram form in order to avoid obscuring the concepts of the described examples.

(259) The description herein is provided to enable a person having ordinary skill in the art to make or use the disclosure. Various modifications to the disclosure will be apparent to a person having

ordinary skill in the art, and the generic principles defined herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein, but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

Claims

1. A method for wireless communications at a user equipment (UE), comprising: identifying that the UE is in communication with a primary cell and one or more secondary cells in a carrier aggregation configuration, wherein the primary cell is a cross-carrier scheduled cell and a first secondary cell is a cross-carrier scheduling cell; monitoring respective UE-specific search spaces of the one or more secondary cells for control information in accordance with a shared search space monitoring configuration; determining a limit corresponding to the shared search space monitoring configuration, wherein the limit is associated with a maximum number of blind decodes to be attempted by the UE, a maximum number of non-overlapping control channel elements, or both, across control channel candidates of the UE-specific search space of the first secondary cell; receiving an indication of the limit from the first secondary cell, wherein determining the limit is based at least in part on receiving the indication; receiving one or more downlink control information messages as a result of monitoring the UE-specific search space of the first secondary cell, wherein the one or more downlink control information messages schedule one or more data transmissions and include a cross-carrier scheduling indication that the one or more data transmissions are scheduled with the primary cell; and communicating with the primary cell via the data transmission in accordance with at least one of the one or more downlink control information messages.
2. The method of claim 1, further comprising: identifying a respective carrier indicator field in each of the one or more downlink control information messages, wherein the cross-carrier scheduling indication is a value included in one of the respective carrier indicator fields.
3. The method of claim 1, further comprising: receiving an indication of the shared search space monitoring configuration consistent with the limit, wherein the monitoring is further in accordance with the limit.
4. The method of claim 1, further comprising: receiving an indication of the shared search space monitoring configuration, wherein the shared search space monitoring configuration includes a number of blind decodes that exceeds the maximum number of blind decodes associated with the limit, a number of non-overlapping control channel elements that exceeds the maximum number of non-overlapping control channel elements associated with the limit, or both; identifying a second UE-specific search space of the first secondary cell included in the shared search space monitoring configuration; and refraining from monitoring the second UE-specific search space of the first secondary cell in accordance with the limit.
5. The method of claim 1, further comprising: receiving a second downlink control information message as a result of the monitoring, wherein the second downlink control information message includes a second scheduling indication that a second data transmission is scheduled with the first secondary cell; and communicating with the first secondary cell via the second data transmission in accordance with the second downlink control information message.
6. The method of claim 1, further comprising: receiving a third downlink control information message as a result of the monitoring, wherein the third downlink control information message includes a third scheduling indication that a third data transmission is scheduled with the primary cell; and communicating with the primary cell via the third data transmission in accordance with the third downlink control information message.
7. The method of claim 1, wherein: the shared search space monitoring configuration includes control channel candidates associated with the one or more downlink control information messages

that include the cross-carrier scheduling indication; and the control channel candidates associated with the one or more downlink control information messages are not included in UE-specific search spaces unassociated with the cross-carrier scheduling indication.

8. The method of claim 1, wherein: the shared search space monitoring configuration includes control channel candidates associated with the one or more downlink control information messages that include the cross-carrier scheduling indication; and one or more of the control channel candidates associated with the one or more downlink control information messages are included in a second UE-specific search space not associated with the cross-carrier scheduling indication.

9. The method of claim 1, further comprising: reporting an indication of a capability of the UE to support shared search space control channel candidates in UE-specific search spaces not associated with the cross-carrier scheduling indication, wherein the shared search space monitoring configuration is based at least in part on the capability.

10. The method of claim 1, wherein communicating with the primary cell via the data transmission comprises: receiving a downlink data transmission from the primary cell, transmitting an uplink data transmission to the primary cell, or both.

11. An apparatus for wireless communications at a user equipment (UE), comprising: one or more processors, one or more memories coupled with the one or more processors; and instructions stored in the one or more memories and executable by the one or more processors to cause the apparatus to: identify that the UE is in communication with a primary cell and one or more secondary cells in a carrier aggregation configuration, wherein the primary cell is a cross-carrier scheduled cell and a first secondary cell is a cross-carrier scheduling cell; monitor respective UE-specific search spaces of the one or more secondary cells for control information in accordance with a shared search space monitoring configuration; determine a limit corresponding to the shared search space monitoring configuration, wherein the limit is associated with a maximum number of blind decodes to be attempted by the UE, a maximum number of non-overlapping control channel elements, or both, across control channel candidates of the UE-specific search space of the first secondary cell; receive an indication of the limit from the first secondary cell, wherein determining the limit is based at least in part on receiving the indication; receive one or more downlink control information messages as a result of monitoring the UE-specific search space of the first secondary cell, wherein the one or more downlink control information messages schedule one or more data transmissions and include a cross-carrier scheduling indication that the one or more data transmissions are scheduled with the primary cell; and communicate with the primary cell via the data transmission in accordance with at least one of the one or more downlink control information messages.

12. The apparatus of claim 11, wherein the instructions are further executable by the one or more processors to cause the apparatus to: identify a respective carrier indicator field in each of the one or more downlink control information messages, wherein the cross-carrier scheduling indication is a value included in one of the respective carrier indicator fields.

13. The apparatus of claim 11, wherein the instructions are further executable by the one or more processors to cause the apparatus to: receive a second downlink control information message as a result of the monitoring, wherein the second downlink control information message includes a second scheduling indication that a second data transmission is scheduled with the first secondary cell; and communicate with the first secondary cell via the second data transmission in accordance with the second downlink control information message.

14. The apparatus of claim 11, wherein the instructions are further executable by the one or more processors to cause the apparatus to: receive a third downlink control information message as a result of the monitoring, wherein the third downlink control information message includes a third scheduling indication that a third data transmission is scheduled with the primary cell; and communicate with the primary cell via the third data transmission in accordance with the third downlink control information message.

15. The apparatus of claim 11, wherein: the shared search space monitoring configuration includes

control channel candidates associated with the one or more downlink control information messages that include the cross-carrier scheduling indication; and the control channel candidates associated with the one or more downlink control information messages are not included in UE-specific search spaces unassociated with the cross-carrier scheduling indication.

16. The apparatus of claim 11, wherein: the shared search space monitoring configuration includes control channel candidates associated with the one or more downlink control information messages that include the cross-carrier scheduling indication, and one or more of the control channel candidates associated with the one or more downlink control information messages are included in a second UE-specific search space not associated with the cross-carrier scheduling indication.

17. The apparatus of claim 11, wherein the instructions are further executable by the one or more processors to cause the apparatus to: report an indication of a capability of the UE to support shared search space control channel candidates in UE-specific search spaces not associated with the cross-carrier scheduling indication, wherein the shared search space monitoring configuration is based at least in part on the capability.

18. The apparatus of claim 11, wherein the instructions to communicate with the primary cell via the data transmission are executable by the one or more processors to cause the apparatus to: receive a downlink data transmission from the primary cell, transmit an uplink data transmission to the primary cell, or both.

19. An apparatus for wireless communications at a user equipment (UE), comprising: means for identifying that the UE is in communication with a primary cell and one or more secondary cells in a carrier aggregation configuration, wherein the primary cell is a cross-carrier scheduled cell and a first secondary cell is a cross-carrier scheduling cell; means for monitoring respective UE-specific search spaces of the one or more secondary cells for control information in accordance with a shared search space monitoring configuration; means for determining a limit corresponding to the shared search space monitoring configuration, wherein the limit is associated with a maximum number of blind decodes to be attempted by the UE, a maximum number of non-overlapping control channel elements, or both, across control channel candidates of the UE-specific search space of the first secondary cell; means for receiving an indication of the limit from the first secondary cell, wherein determining the limit is based at least in part on receiving the indication; means for receiving one or more downlink control information messages as a result of monitoring the UE-specific search space of the first secondary cell, wherein the one or more downlink control information messages schedule one or more data transmissions and include a cross-carrier scheduling indication that the one or more data transmissions are scheduled with the primary cell; and means for communicating with the primary cell via the data transmission in accordance with at least one of the one or more downlink control information messages.

20. A non-transitory computer-readable medium storing code for wireless communications at a user equipment (UE), the code comprising instructions executable by one or more processors to: identify that the UE is in communication with a primary cell and one or more secondary cells in a carrier aggregation configuration, wherein the primary cell is a cross-carrier scheduled cell and a first secondary cell is a cross-carrier scheduling cell; monitor respective UE-specific search spaces of the one or more secondary cells for control information in accordance with a shared search space monitoring configuration; determine a limit corresponding to the shared search space monitoring configuration, wherein the limit is associated with a maximum number of blind decodes to be attempted by the UE, a maximum number of non-overlapping control channel elements, or both, across control channel candidates of the UE-specific search space of the first secondary cell; receive an indication of the limit from the first secondary cell, wherein determining the limit is based at least in part on receiving the indication; receive one or more downlink control information messages as a result of monitoring the UE-specific search space of the first secondary cell, wherein the one or more downlink control information messages schedule one or more data transmissions and include a cross-carrier scheduling indication that the one or more data transmissions are

scheduled with the primary cell; and communicate with the primary cell via the data transmission in accordance with at least one of the one or more downlink control information messages.